



INFRATEMPORAL FOSSA

The infratemporal fossa lies deep to the ramus of the mandible. It communicates with the temporal fossa superiorly deep to the zygomatic arch, the orbit anteriorly through the inferior orbital fissure, and the pterygopalatine fossa medially through the pterygomaxillary fissure. It also communicates with the middle fossa through the foramina ovale and spinosum. The major structures that occupy the infratemporal fossa are the lateral and medial pterygoid muscles, the mandibular division of the trigeminal nerve, the chorda tympani branch of the facial nerve, the otic parasympathetic ganglion, the maxillary artery and the pterygoid venous plexus.

The infratemporal fossa has a roof and anterior, lateral and medial walls, and is open to the neck posteroinferiorly, i.e. it has no anatomical floor. Approximately 80% of the roof is formed by the infratemporal surface of the greater wing of the sphenoid. The remainder is formed by the infratemporal surface of the temporal bone, ending at the articular eminence of the temporomandibular joint and the spine of the sphenoid on the deep medial aspect. It contains the foramina ovale and spinosum. The anterior wall is formed by the posterior surface of the maxilla, ending inferiorly at the maxillary tuberosity. The inferior orbital fissure forms the upper limit of the anterior wall, meeting the pterygomaxillary fissure at right angles. The medial wall is formed anteriorly by the lateral pterygoid plate of the pterygoid process of the sphenoid, and more posteromedially by the pharynx and tensor and levator veli palatini. It contains the pterygomaxillary fissure, across which structures pass between the infratemporal and pterygopalatine fossae (Fig. 32.1). The lateral wall is formed by the medial surface of the ramus of the mandible.

Lateral pterygoid provides a key to understanding the relationships of structures within the infratemporal fossa. This muscle lies in the roof of the fossa and runs anteroposteriorly in a more or less horizontal plane from the region of the pterygoid plates to the mandibular condyle (see Fig. 32.1). Branches of the mandibular nerve and the main origin of medial pterygoid are deep relations and the maxillary artery is superficial. The buccal branch of the mandibular nerve passes between the two heads of lateral pterygoid. Medial pterygoid and the lingual and

inferior alveolar nerves emerge below its inferior border, and the deep temporal nerves and vessels emerge from its upper border. A venous network, the pterygoid venous plexus, lies around and within lateral pterygoid and is important in the spread of infection.

BONES

The sphenoid bone, the paired maxillae and temporal bones, the palatine bones and the mandible collectively provide the skeletal framework to the infratemporal and pterygopalatine regions. The mandible and the two temporal bones articulate at the right and left temporomandibular joints. The disarticulated maxilla and palatine bone are described on pages 484 and 486, respectively the temporal bone is described on page 624, and the sphenoid and mandible are described here.

Sphenoid bone

The sphenoid bone lies in the base of the skull between the frontal, temporal and occipital bones. It has a central body, paired greater and lesser wings that spread laterally from the body, and two pterygoid processes that descend from the junction of the body and greater wings (Fig. 32.3).

Body

The body of the sphenoid is cuboidal. It contains two air sinuses, separated by a septum. Its cerebral (superior) surface articulates in front with the cribriform plate of the ethmoid bone. Anteriorly lies the smooth jugum sphenoidale, which is related to the gyri recti and olfactory tracts. The jugum is bounded behind by the anterior border of the sulcus chiasmaticus, which leads laterally to the optic canals. Posteriorly lies the tuberculum sellae, behind which is the deeply concave sella turcica. In life, the sella contains the hypophysis cerebri in the hypophysial fossa. Its anterior edge is completed laterally by two middle clinoid processes, while posteriorly the sella turcica is bounded by a square dorsum sellae, the superior angles of which bear variable posterior clinoid processes. The diaphragma sella and the tentorium cerebelli are attached to the clinoid processes. On each side, below the

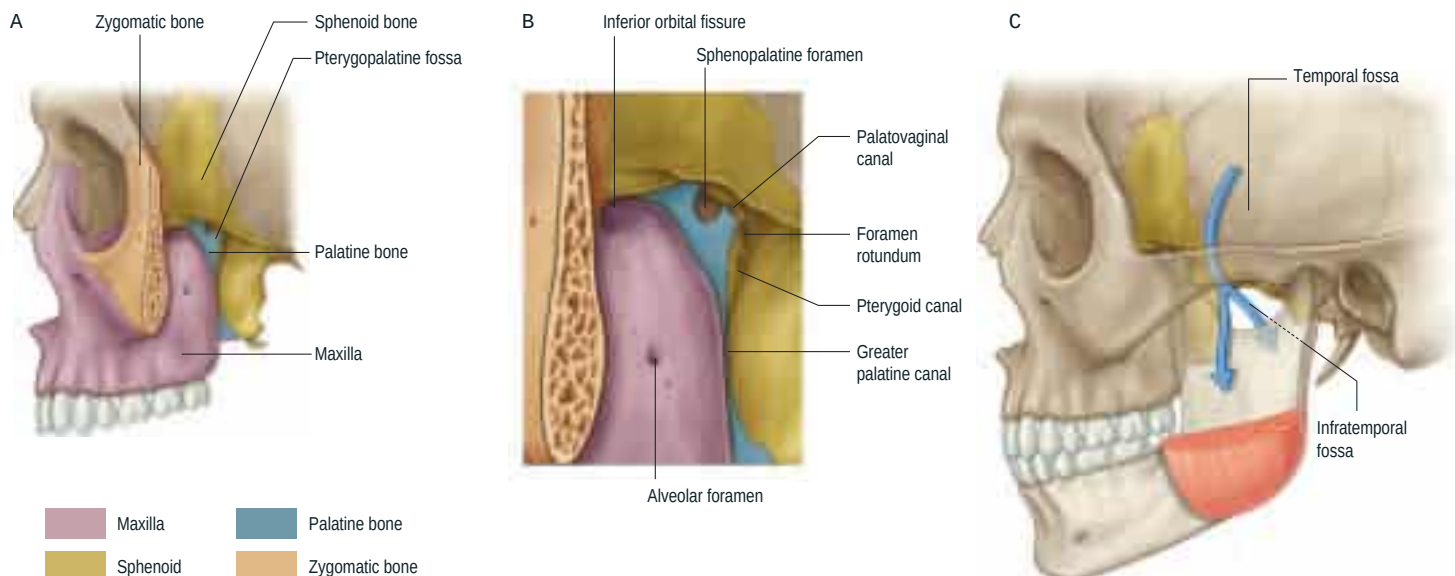


Fig. 32.1 **A**, The pterygopalatine fossa after detachment of the mandible and removal of the zygomatic arch. **B**, Anteromedial view. **C**, Lateral view; the arrows show that the floor of the temporal fossa is open medially to the infratemporal fossa and laterally to the region containing the masseter. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

There is a lack of consensus in the surgical literature with regard to both the boundaries and contents of the infratemporal fossa. Thus, the fossa is sometimes defined as the anatomical space beneath the floor of the middle fossa, incorporating the remainder of the subcranial temporal bone as part of the roof, with the exception of the glenoid fossa of the temporomandibular joint. In this description, the fossa is limited posteriorly by the prevertebral fascia and includes the internal carotid artery, the internal jugular vein, the lower cranial nerves, the cervical sympathetic trunk, and the styloid process with its attached muscles and ligaments. Alternatively, the 'extended' infratemporal fossa is said to consist of the parapharyngeal and masticator spaces (Fig. 32.2) (Falcon et al 2010, Hosseini et al 2012, Bejjani et al 1998, Joo et al 2013).

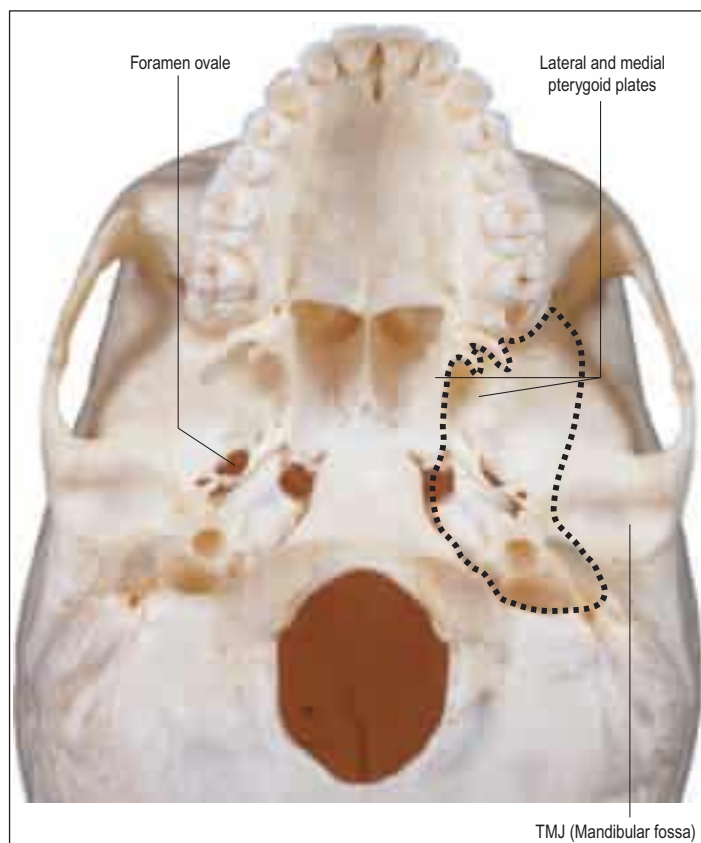


Fig. 32.2 The 'extended' infratemporal fossa. The boundaries of the infratemporal fossa are delineated by the dotted line. Note that the lateral pterygoid plate and pharyngotympanic tube constitute the medial wall of the fossa, and that the temporomandibular joint (TMJ) forms part of its lateral wall. The carotid and jugular foramina lie in the posterior part of this extended infratemporal fossa. (Adapted with permission from Berkovitz BKB, Moxham BJ, Telling D 1994 *Color Atlas of the Skull*, London, Mosby and Hosseini SMS, Razfar A, Carrau RL, et al. Endonasal transpterygoid approach to the infratemporal fossa: correlation of endoscopic and multiplanar CT anatomy. *Head and Neck*. 2012;34:313–320.)

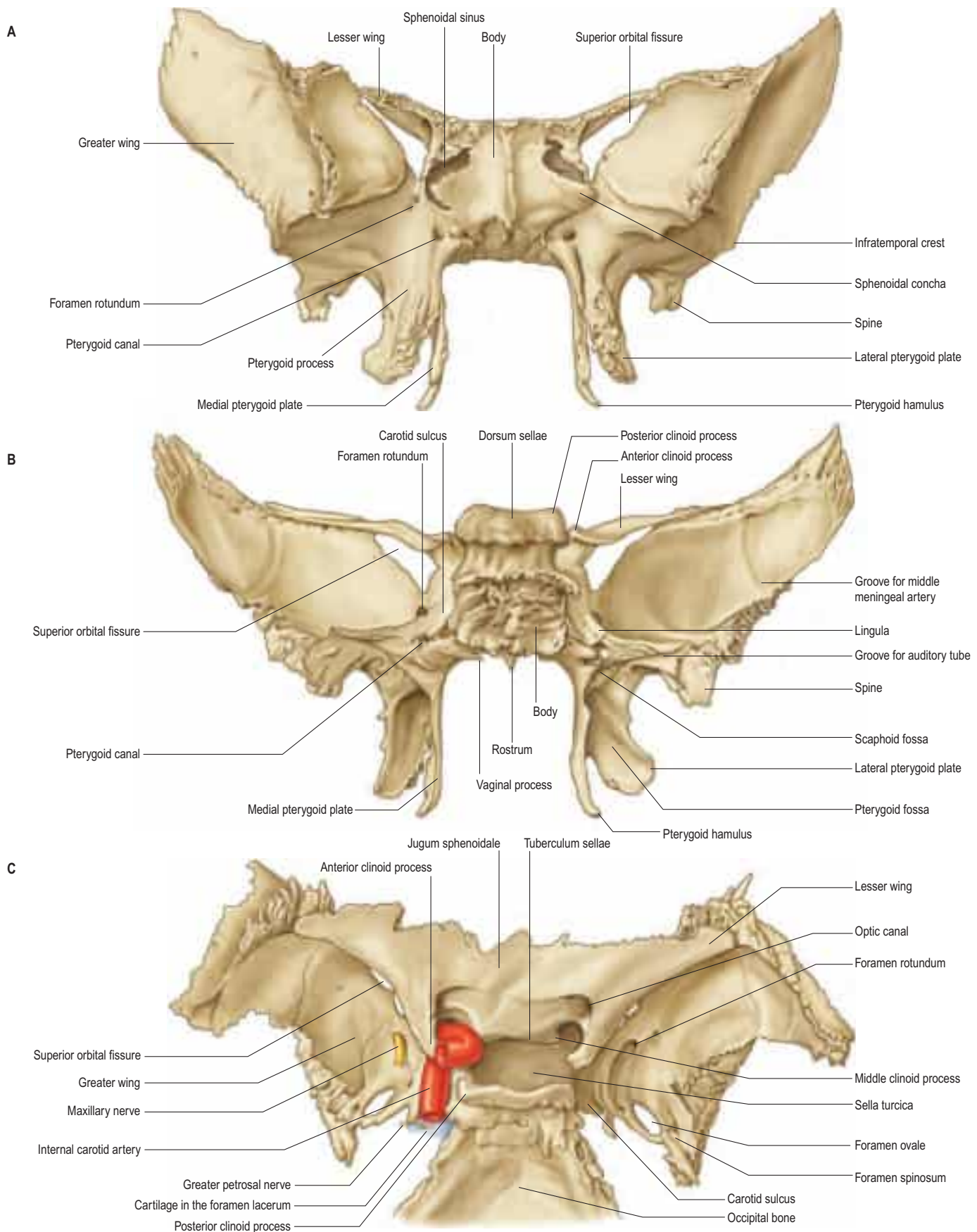


Fig. 32.3 The sphenoid bone. **A**, Anterior view. **B**, Posterior view. **C**, Superior view. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

dorsum sellae, a small petrosal process articulates with the apex of the petrous part of the temporal bone. The body of the sphenoid slopes directly into the basilar part of the occipital bone posterior to the dorsum sellae; together these bones form the clivus. In the growing child, this is the site of the spheno-occipital synchondrosis; premature closure of this joint gives rise to the skull appearances seen in achondroplasia.

The lateral surfaces of the body are united with the greater wings and the medial pterygoid plates. A broad carotid sulcus accommodates both the internal carotid artery and the cranial nerves associated with the cavernous sinus above the root of each wing. The sulcus is deepest posteriorly. It is overhung medially by the petrosal part of the temporal bone and has a sharp lateral margin, the lingula, which continues back over the posterior opening of the pterygoid canal.

A median triangular, bilaminar sphenoidal crest on the anterior surface of the body of the sphenoid makes a small contribution to the nasal septum. The anterior border of the crest joins the perpendicular plate of the ethmoid bone, and a sphenoidal sinus opens on each side of it. In the articulated state, the sphenoidal sinuses are closed antero-inferiorly by the sphenoidal conchae, which are largely destroyed when disarticulating a skull. Each half of the anterior surface of the body of the sphenoid possesses a superolateral depressed area joined to the ethmoidal labyrinth that completes the posterior ethmoidal sinuses; a lateral margin that articulates with the orbital plate of the ethmoid above and the orbital process of the palatine bone below; and an inferomedial, smooth, triangular area, which forms the posterior nasal roof, and near whose superior angle lies the orifice of a sphenoidal sinus.

The inferior surface of the body of the sphenoid bears a median triangular sphenoidal rostrum, embraced above by the diverging lower margins of the sphenoidal crest. The narrow anterior end of the rostrum fits into a fissure between the anterior parts of the alae of the vomer, and the posterior ends of the sphenoidal conchae flank the rostrum, articulating with its alae. A thin vaginal process projects medially from the base of the medial pterygoid plate on each side of the posterior part of the rostrum, behind the apex of the sphenoidal concha.

Greater wings

The greater wings of the sphenoid curve broadly superolaterally from the body. Posteriorly, each is triangular, fitting the angle between the petrous and squamous parts of the temporal bone at a sphenosquamosal suture. The cerebral surface contributes to the anterior part of the middle cranial fossa. Deeply concave, its undulating surface is adapted to the anterior gyri of the temporal lobe of the cerebral hemisphere. The foramen rotundum lies anteromedially and transmits the maxillary nerve. Posterolateral to the foramen rotundum is the foramen ovale, which transmits the mandibular nerve, accessory meningeal artery and sometimes the lesser petrosal nerve, although the latter nerve may have its own canaliculus innominatus medial to the foramen spinosum. A small emissary sphenoidal foramen (foramen of Vesalius), which transmits a small vein from the cavernous sinus, sometimes lies medial to the foramen ovale (on one or both sides). The foramen spinosum, which transmits the middle meningeal artery and meningeal branch of the mandibular nerve, lies behind the foramen ovale.

The lateral surface is vertically convex and divided by a transverse infratemporal crest into temporal (upper) and infratemporal (lower) surfaces. Temporalis is attached to the temporal surface. The infratemporal surface is directed downwards and, with the infratemporal crest, is the site of attachment of the upper fibres of lateral pterygoid. It contains the foramen ovale and foramen spinosum. The small downward-projecting spine of the sphenoid lies posterior to the foramen spinosum; the sphenomandibular ligament is attached to its tip. The medial side of the spine bears a faint antero-inferior groove for the chorda tympani nerve and appears in the lateral wall of the sulcus for the pharyngotympanic (auditory) tube. Medial to the anterior end of the infratemporal crest, a ridge passes downwards to the front of the lateral pterygoid plate, thereby forming a posterior boundary of the pterygomaxillary fissure.

The quadrilateral orbital surface of the greater wing faces antero-medially and forms the posterior part of the lateral wall of the orbit. It has a serrated upper edge, which articulates with the orbital plate of the frontal bone, and a serrated lateral margin, which articulates with the zygomatic bone. Its smooth inferior border is the posterolateral edge of the inferior orbital fissure, and its sharp medial margin forms the inferolateral edge of the superior orbital fissure, on which a small tubercle gives partial attachment to the common annular ocular tendon. Below the medial end of the superior orbital fissure, a grooved area forms the posterior wall of the pterygopalatine fossa; the latter is pierced by the foramen rotundum.

The irregular margin of the greater wing, from the body of the sphenoid to the spine, is an anterior limit of the medial half of the foramen lacerum. It also displays the posterior aperture of the pterygoid canal. Its lateral half articulates with the petrous part of the temporal bone at a sphenopetrosal synchondrosis. Inferior to this, the sulcus tubae contains the cartilaginous pharyngotympanic (auditory) tube. Anterior to the spine of the sphenoid the concave squamosal margin is serrated – bevelled internally below, externally above – for articulation with the squamous part of the temporal bone. The tip of the greater wing, bevelled internally, articulates with the sphenoidal angle of the parietal bone at the pterion. Medial to this, a triangular rough area articulates with the frontal bone; its medial angle is continuous with the inferior boundary of the superior orbital fissure, and its anterior angle joins the zygomatic bone by a serrated articulation. Fractures of the zygomatic

bone preferentially pass through this articulation with the greater wing because it is a line of weakness in the lateral orbit.

Lesser wings

The lesser wings of the sphenoid are triangular pointed plates that protrude laterally from the anterosuperior regions of the body. The superior surface of each wing is smooth and related to the frontal lobe of the cerebral hemisphere. The inferior surface is a posterior part of the orbital roof and upper boundary of the superior orbital fissure, and overhangs the middle cranial fossa. The posterior border projects into the lateral fissure of the cerebral hemisphere. The medial end of the lesser wing forms the anterior clinoid process. The anterior and middle clinoid processes are sometimes united to form a caroticoclinoid foramen. The lesser wing is connected to the body by a thin, flat anterior root and a thick, triangular posterior root (the optic strut), between which lies the optic canal. The optic strut extends from the base of the anterior clinoid process to the body and separates the optic canal from the superior orbital fissure. The optic canal is bounded by the body of the sphenoid medially, the lesser wing superiorly, and the optic strut inferiorly and laterally.

Superior orbital fissure

The superior orbital fissure connects the cranial cavity with the orbit. It is bounded medially by the body of the sphenoid, above by the lesser wing of the sphenoid, below by the medial margin of the orbital surface of the greater wing, and laterally, between the greater and lesser wings, by the frontal bone. The contents of the superior orbital fissure are described on p. 667 (see Fig. 41.4).

Pterygoid processes

The pterygoid processes descend perpendicularly from the junctions of the greater wings and body. Each consists of a medial and lateral plate, whose upper parts are fused anteriorly. The plates are separated below by the angular pterygoid fissure, whose margins articulate with the pyramidal process of the palatine bone, and diverge behind. Medial pterygoid lies in the cuneiform pterygoid fossa between the plates. Above is the small, oval, shallow scaphoid fossa, which is formed by division of the upper posterior border of the medial plate. Part of tensor veli palatini is attached to the fossa. The anterior surface of the root of the pterygoid process is broad and triangular, and forms the posterior wall of the pterygopalatine fossa; it is pierced by the anterior opening of the pterygoid canal.

Lateral pterygoid plate

The lateral pterygoid plate is broad, thin and everted. The lateral surface forms part of the medial wall of the infratemporal fossa; the lower part of lateral pterygoid is attached to it. The medial surface is the lateral wall of the pterygoid fossa; most of the deep head of medial pterygoid is attached to it. The upper part of its anterior border is a posterior boundary of the pterygomaxillary fissure, and the lower part articulates with the palatine bone. The posterior border is free.

Medial pterygoid plate

The medial pterygoid plate is narrower and longer than the lateral. Its lower end is continued into an unciform projection, the pterygoid hamulus, which curves laterally. The pterygomandibular raphe is attached to the hamulus, and the tendon of tensor veli palatini winds around the hamulus. The lateral surface forms the medial wall of the pterygoid fossa and the medial surface provides a lateral boundary of the posterior nasal aperture. The medial plate is prolonged above on the inferior aspect of the body of the sphenoid as a thin vaginal process that articulates anteriorly with the sphenoidal process of the palatine bone and medially with the ala of the vomer. The plate articulates with the posterior border of the perpendicular plate of the palatine bone in the lower part of its anterior margin. Inferiorly, it bears a furrow, which is converted anteriorly into the palatovaginal canal by the sphenoidal process of the palatine bone. The palatovaginal canal transmits pharyngeal branches of the maxillary artery and pterygopalatine ganglion. The pharyngobasilar fascia is attached to the whole of the posterior margin of the medial plate, and the superior pharyngeal constrictor is attached to its lower end. The small pterygoid tubercle is found at the upper end of the plate, just below the posterior opening of the pterygoid canal. The processus tubarius, which supports the cartilaginous pharyngeal end of the pharyngotympanic tube, projects back near the midpoint of the margin of the medial pterygoid plate.

Though adjacent, the medial and lateral pterygoid plates have distinct roles. The lateral plate is the province of the infratemporal fossa and masticator space, forming part of the medial boundary of the infratemporal fossa and providing origin to the lateral and medial

pterygoids. The medial plate is functionally related to the pharynx, providing attachment for the pharyngobasilar fascia, superior constrictor and pterygomandibular raphe.

Midface fractures Midface fractures at Le Fort I, II and III levels invariably involve the pterygoid plates; displacement of the outlines of these bones radiographically confirms the diagnosis.

Sphenoidal conchae

The sphenoidal conchae are two thin, curved small plates, attached anteroinferiorly to the body of the sphenoid bone. The superior concave surface of each forms the anterior wall and part of the floor of a sphenoidal sinus. *In situ*, each has vertical quadrilateral anterior and horizontal triangular posterior parts. The anterior part consists of a superolateral depressed area, which completes the posterior ethmoidal sinuses and joins below with the orbital process of a palatine bone, and a smooth and triangular inferomedial area, which forms part of the nasal roof and is perforated above by the round opening connecting the sphenoidal sinus and sphenoid-ethmoidal recess.

Anterior parts of the two bones meet in the midline, and protrude as the sphenoidal crest. The horizontal part appears in the nasal roof and completes the sphenopalatine foramen. Its medial edge articulates with the rostrum of the sphenoid and the ala of the vomer. Its apex, directed posteriorly, is superomedial to the vaginal process of the medial pterygoid plate and joins the posterior part of the ala. A small conchal part sometimes appears in the medial wall of the orbit, lying between the orbital plate of the ethmoid in front, the orbital process of the palatine bone below and the frontal bone above.

Ossification

Until the seventh or eighth month *in utero*, the sphenoid body has a presphenoidal part, anterior to the tuberculum sellae, with which the lesser wings are continuous, and a postsphenoidal part, consisting of the sella turcica and dorsum sellae, and integral with the greater wings and pterygoid processes. Much of the bone is preformed in cartilage. There are six ossification centres for the presphenoidal parts and eight for the postsphenoidal parts.

Presphenoidal part

At about the ninth week of fetal life, a centre appears in each wing, lateral to the optic canal, and a little later two bilateral centres appear in the presphenoidal body. Each sphenoidal concha has a centre, appearing superoposteriorly in the nasal capsule in the fifth month *in utero*. As this enlarges, it partly surrounds a posterosuperior expansion of the nasal cavity, which becomes the sphenoidal sinus. The posterior conchal wall is absorbed and the sinus invades the presphenoidal component. In the fourth year, the concha fuses with the ethmoidal labyrinth and, before puberty, it fuses with the sphenoid and palatine bones. Its anterior deficiency persists as an opening for the sphenoidal sinus.

Postsphenoidal part

The first centres appear in the greater wings at about the eighth week of fetal life, one in the basal cartilage of each wing below the foramen rotundum. These centres only contribute to the root of the greater wing (near the foramen rotundum and pterygoid canal). The remainder of the greater wing and the lateral pterygoid plate are ossified in mesenchyme. At about the fourth month of fetal life, two centres appear, flanking the sella turcica, and soon fuse. The medial pterygoid plates are also ossified in 'membrane', a centre in each probably appearing in about the ninth or tenth week. The hamulus is chondrified during the third fetal month and at once begins to ossify. Medial and lateral pterygoid plates join at about the sixth fetal month. During the fourth month, a centre appears for each lingula, soon joining the body. The optic canal in the neonate is relatively large and has a keyhole or 'figure of eight' shape rather than the circular profile seen in the adult (Lang 2001).

Postnatal details

Presphenoidal and postsphenoidal parts fuse at about the eighth month *in utero* but an unciform cartilage persists after birth in lower parts of the junction. At birth, the bone is tripartite and consists of a central part (body and lesser wings) and two lateral parts (each consisting of a greater wing and pterygoid process). During the first year, the greater wings and body unite around the pterygoid canals, and the lesser wings extend medially above the anterior part of the body, meeting to form the smooth, elevated jugum sphenoidale. By the twenty-fifth year, sphenoid and occipital bones are completely fused. An occasional vascular foramen, often erroneously termed the craniopharyngeal canal, is occasionally seen in the anterior part of the hypophysial fossa.

Although the sphenoidal sinus can be identified in the fourth month of fetal life as an evagination of the posterior part of the nasal capsule, by birth it represents an outgrowth of the sphenoid-ethmoidal recess. Pneumatization of the body of the sphenoid commences at around 7 months of age and a distinct cell is visible by the age of 2 years. Pneumatization spreads first into the presphenoid and later invades the postsphenoid part. The sinus reaches full size in adolescence, but often enlarges further by absorption of its walls as age advances.

Certain parts of the sphenoid are connected by ligaments that may occasionally ossify, e.g. the pterygospinous ligament between the sphenoidal spine and the upper part of the lateral pterygoid plate; the interclinoid ligament joining the anterior to the posterior clinoid process; and the caroticoclinoid ligament that connects the anterior to the middle clinoid process.

Premature synostosis of the junction between pre- and postsphenoidal parts, or of the sphenoid-occipital suture, produces a characteristic appearance, obvious in profile, of an abnormal depression of the nasal bridge.

Stylohamular plane



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Mandible

The mandible is the largest, strongest and lowest bone in the face. It has a horizontally curved body that is convex forwards, and two broad rami that ascend posteriorly (Fig. 32.5). The body of the mandible supports the mandibular teeth within the alveolar process. The rami bear the coronoid and condylar processes. Each condyle articulates with the adjacent temporal bone at the temporomandibular joint.

Body

The body is somewhat U-shaped. It has external and internal surfaces separated by upper and lower borders. Anteriorly, the upper external surface shows an inconstant faint median ridge indicating the site of the fused symphysis menti. Inferiorly, this ridge divides to enclose a triangular mental protuberance; its base is centrally depressed but raised on each side as a mental tubercle. The mental protuberance and mental tubercles constitute the chin. The mental foramen, from which the mental neurovascular bundle emerges, lies below either the interval between the premolar teeth, or the second premolar tooth, midway between the upper and lower borders of the body. The posterior border of the foramen is smooth and accommodates the nerve as it emerges posterolaterally. A faint external oblique line ascends backwards from each mental tubercle, and sweeps below the mental foramen; it becomes more marked as it continues into the anterior border of the ramus.

The lower border of the body, the base, extends posterolaterally from the mandibular symphysis into the lower border of the ramus behind the third molar tooth. Near the midline on each side, there is a rough digastric fossa, which gives attachment to the anterior belly of digastric. Behind the fossa, the base is thick and rounded: it has a slight anteroposterior convexity that changes to a gentle concavity as the ramus is approached, and so the base has an overall sinuous profile.

The upper border, the alveolar part, contains 16 alveoli for the roots of the lower teeth. It consists of buccal and lingual plates of bone joined by interdental and inter-radicular septa. Near the second and third molar teeth, the external oblique line is superimposed on the buccal plate. As in the maxilla, the form and depth of the tooth sockets is related to the morphology of the roots of the mandibular teeth. The sockets of the incisor, canine and premolar teeth usually contain a single root, while those for the three molar teeth each contain two or three roots. The third molar is variable in its position and root presentation. It may be impacted vertically, horizontally, mesially or distally, and its roots may be bulbous, hooked, divergent or convergent, and occasionally embrace the mandibular (inferior dental) canal. The internal surface of the mandible is divided by an oblique mylohyoid line that gives attachment to mylohyoid (and, above its posterior end, to the superior pharyngeal constrictor, some retromolar fascicles of buccinator, and the pterygomandibular raphe behind the third molar). The mylohyoid line extends from a point approximately 1 cm from the upper border behind the third molar as far forwards as the mental symphysis; it is sharp and distinct near the molars but faint further forwards. The mylohyoid groove extends downwards and forwards from the ramus below the posterior part of the mylohyoid line and contains the mylohyoid neurovascular bundle. The area below the line is a slightly concave submandibular fossa and is related to the submandibular gland. The area above the line widens anteriorly into a triangular

A plane exists between the pharyngeal side wall and the infratemporal fossa/masticator space. It is readily identified by palpating the tip of the styloid process and with gentle blunt finger dissection passing anteriorly and medially to the pterygoid hamulus at the lower border of the medial pterygoid plate. Termed the 'stylohamular plane' (Friedman et al 1981), it defines the medial and superior boundary of the infratemporal fossa and is the medial boundary for tumours confined to the infratemporal fossa. With resections using the stylohamular plane as a guide, the internal carotid, internal jugular vein and the vagus nerve should not be at risk because they are deep to the styloid apparatus (Fig. 32.4).

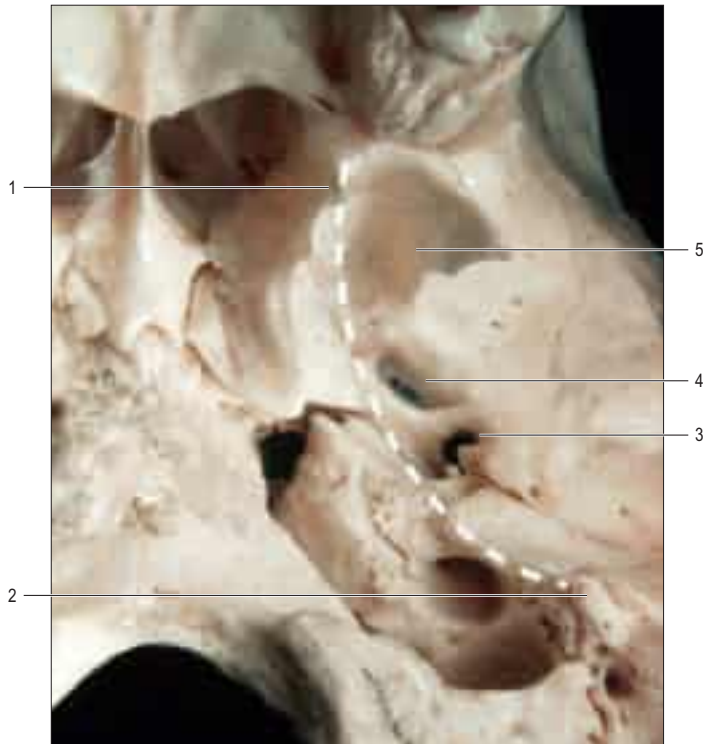


Fig. 32.4 The stylohamular plane (dotted line). Key: 1, pterygoid hamulus; 2, styloid process; 3, foramen spinosum; 4, foramen ovale; 5, lateral pterygoid plate.

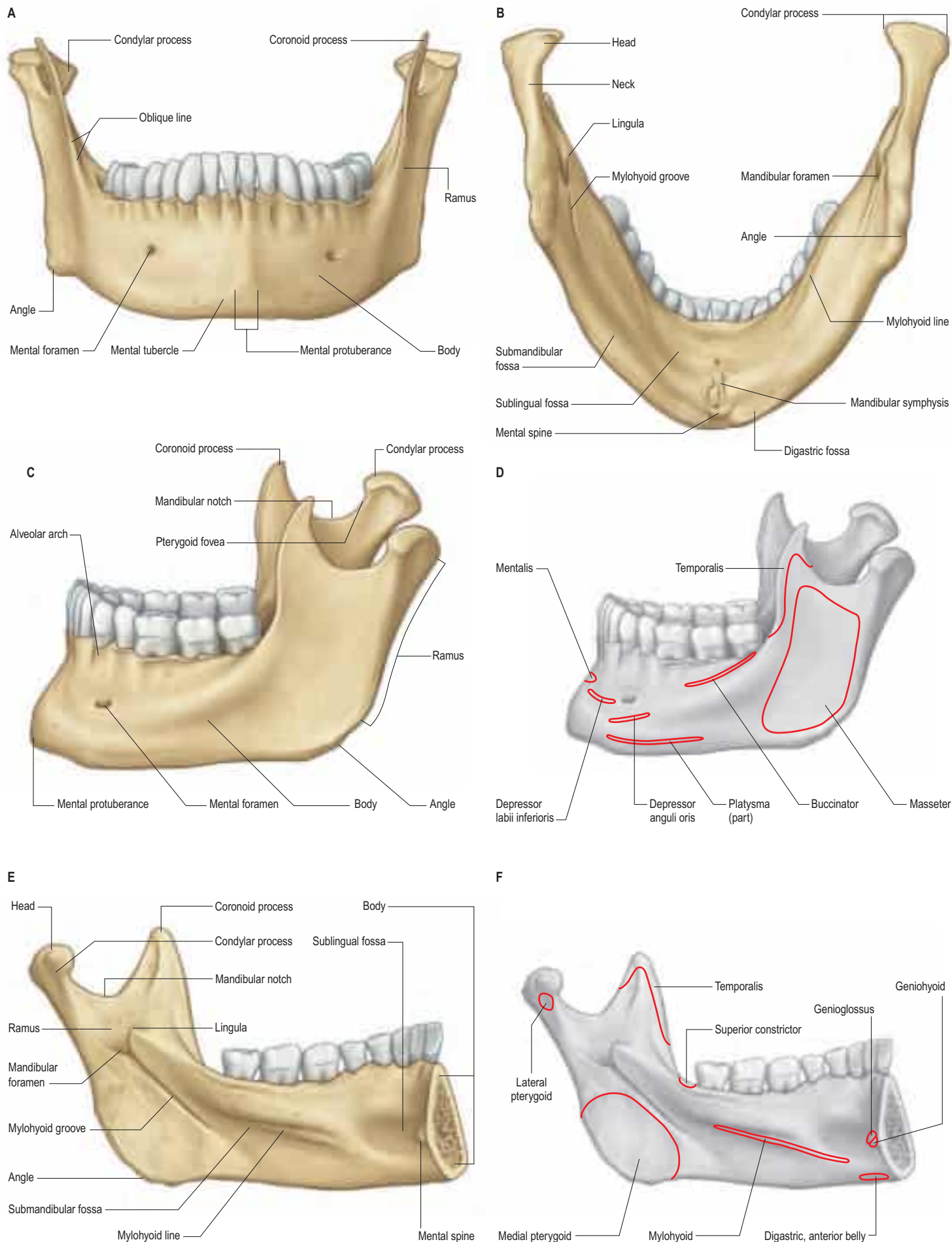


Fig. 32.5 The adult mandible. **A**, Anterior view. **B**, Inferior view. **C**, Lateral view (with muscle attachments in **D**). **E**, Internal view (with muscle attachments in **F**). (Redrawn with permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

sublingual fossa and is related to the sublingual gland; the bone is covered by mucosa above the sublingual fossa as far back as the third molar. In an edentulous subject, it may be necessary to reduce any ridge-like prominence of the mylohyoid line in order for dentures to fit without traumatizing the overlying oral mucosa.

Above the anterior ends of the mylohyoid lines, the posterior symphyseal aspect bears a small elevation, often divided into upper and lower parts, the mental spines (genial tubercles). The spines are sometimes fused to form a single eminence, or they may be absent, in which case their position is indicated merely by an irregularity of the surface. The upper part gives attachment to genioglossus, the lower part to geniohyoid. Midline lingual (genial) foramina, above and/or below the genial tubercles, and lateral lingual foramina in the premolar region are present in most mandibles and are of importance in the vascular supply to the symphysis. A rounded elevation of compact bone, the torus mandibularis, sometimes develops above the mylohyoid line; it is most prominent in the premolar region, is usually bilateral and only of clinical significance if repeatedly traumatized.

Ramus

The mandibular ramus is quadrilateral, and has two surfaces (lateral and medial), four borders (superior, inferior, anterior and posterior) and two processes (coronoid and condylar). The lateral surface is relatively featureless and bears the (external) oblique ridge in its lower part. The mandibular foramen, through which the inferior alveolar neurovascular bundle passes to gain access to the mandibular canal (see below), is sited midway between the anterior and posterior borders of the ramus about level with the occlusal surfaces of the teeth. It is overlapped anteromedially by a thin, sharp, triangular spine, the lingula, to which the sphenomandibular ligament is attached, and which is also the landmark for an inferior alveolar local anaesthetic block injection. Below and behind the foramen, the mylohyoid groove runs obliquely downwards and forwards.

The inferior border is continuous with the mandibular base and meets the posterior border at the angle, which is typically everted in males but frequently inverted in females. The thin superior border bounds the mandibular incisure (sigmoid or mandibular notch, incisura semilunaris), which is surmounted in front by the somewhat triangular, flat coronoid process and behind by the condylar process. The thick, rounded posterior border extends from the condyle to the angle, and is gently convex backwards above and concave below. The anterior border is thin above, where it is continuous with the edge of the coronoid process, and thicker below, where it is continuous with the external oblique line. The temporal crest is a ridge that descends on the medial side of the coronoid process from its tip to the bone just behind the third molar tooth. The triangular depression between the temporal crest and the anterior border of the ramus is the retromolar fossa (retromolar trigone).

The ramus and its processes provide attachment for the four primary muscles of mastication. Masseter is attached to the lateral surface, medial pterygoid is attached to the medial surface, temporalis is inserted into the coronoid process and lateral pterygoid is attached to the condyle. The thickness of the ramus decreases markedly behind the external oblique line laterally, the temporal crest medially, and above the lingula; there may be fusion of the lateral and medial cortical plates. This is of importance in mandibular ramus osteotomies and also influences the frequency of fractures of the mandibular angle.

Antilingula



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Mandibular canal

The mandibular foramen leads into the mandibular canal, which runs downwards and forwards within the ramus, gently curving inferiorly within the body under the roots of the molar teeth, with which it communicates by small openings, and ascending in the premolar region to the mental foramen. The canal is not always easy to define on plain X-rays, especially anterior to the mental foramen. Its walls may be formed either by a thin layer of cortical bone or, more frequently, by trabecular bone. Although the buccal-lingual and superior-inferior positions of the canal vary considerably between mandibles, in general, the mandibular canal is situated nearer the lingual cortical plate in the posterior two-thirds of the bone, and closer to the labial cortical plate in the anterior third. Bilateral symmetry (location of the canal in each half of the mandible) is reported to be common. Near the mental foramen, the inferior alveolar nerve branches into the mental nerve, which ultimately leaves the mandible via the mental foramen, and the incisive nerve, which remains within the bone and supplies the anterior

teeth. The mental nerve may extend anteriorly for 2–3 mm within the mandible before curving back to the mental foramen (the 'anterior loop' of the mental nerve). Appreciation of the three-dimensional course of the mandibular canal during its passage through the mandible from the mandibular to the mental foramina is essential if damage to the inferior alveolar nerve is to be avoided in third molar surgery, mandibular osteotomies, dental implant surgery and harvesting of mandibular bone grafts.

Coronoid process

The coronoid process projects upwards and slightly forwards as a triangular plate of bone. Its posterior border bounds the mandibular incisure, and its anterior border continues into that of the ramus.

Coronoid hyperplasia



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Condylar process

The mandibular condyle varies considerably in both size and shape. When viewed from above, the condyle is roughly ovoid in outline, its anteroposterior dimension (approximately 1 cm) being roughly half its mediolateral dimension. The long axis of the condyle is at right angles to the mandibular ramus but, due to the flare of the ramus, the lateral pole of the condyle is slightly anterior to the medial; if the long axes of the two condyles are extended, they meet at an obtuse angle, varying from 145° to 160°, at the anterior border of the foramen magnum. The slender neck of the condyle, which expands transversely upwards, joins the ramus to the articular head. The pterygoid fovea, a small depression situated on the anterior surface of the neck below the articular surface, receives part of the attachment of lateral pterygoid.

The condyle consists of a core of cancellous bone covered by a thin outer layer of compact bone whose intra-articular aspect is covered by layers of fibrocartilage. The condyle is one of the most common sites for mandibular fractures.

Condylar hyperplasia



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Accessory foramina of the mandible

Accessory mandibular foramina are usually unnamed and infrequently described, and yet they are numerous. They may transmit auxiliary nerves to the teeth (from facial, mylohyoid, buccal, transverse cervical cutaneous, lingual and other nerves), and their occurrence is significant in dental anaesthetic blocking techniques. The innervation of the teeth is discussed on [page 526](#). The accessory lingual foramina in the mandibular symphysis are particularly relevant in dental implant surgery and osteotomies, e.g. genioplasty, at this site (see below).

Ossification

The mandible forms in dense fibromembranous tissue lateral to the inferior alveolar nerve and its incisive branch, and also in the lower parts of Meckel's cartilage (first branchial arch). Each half is ossified from a centre that appears near the mental foramen at about the sixth week *in utero*. From this site, ossification spreads medially and posterocranially to form the body and ramus, first below, and then around, the inferior alveolar nerve and its incisive branch. Ossification then spreads upwards, initially forming a trough, and later crypts, for the developing teeth. By the tenth week, Meckel's cartilage below the incisor rudiments is surrounded and invaded by bone. The mandibular condyle and coronoid process do not develop from the primary (Meckel's) cartilage but from secondary cartilages that appear later ([Fig. 32.6](#)). A conical mass, the condylar cartilage, extends from the mandibular head downwards and forwards in the ramus, and contributes to its growth in height. Although it is largely replaced by bone by midfetal life, its proximal end persists as proliferating cartilage under the fibrous articular lining until about the third decade. Another secondary cartilage, which soon ossifies, appears along the anterior coronoid border, and disappears before birth. One or two cartilaginous nodules also occur at the symphysis menti. At about the seventh month *in utero*, these may ossify as variable mental ossicles in symphyseal fibrous tissue; they unite with adjacent bone before the end of the first postnatal year.

Age changes in the mandible

At birth, the two halves of the mandible are united by a fibrous symphysis menti ([Fig. 32.7B](#)). The anterior ends of both rudiments are covered by cartilage, separated only by a symphysis. Until fusion occurs, new cells are added to each cartilage from symphyseal fibrous tissue,

The term 'antilingula' has been used to describe an inconstant bony prominence on the lateral aspect of the ramus. This localized bulge, initially thought to be the result of the entrance of the inferior alveolar neurovascular bundle into the medial ramus, has been used as a guide to the position of the mandibular foramen in mandibular osteotomies. However, there is little correlation between the position of the antilingula and the mandibular foramen. It has been suggested that the prominence (when present) on the lateral ramus could reflect the insertion of fibres of masseter ([Lang 1995](#), [Hogan and Ellis 2006](#), [Park 2014](#)).

The availability of computed tomography (CT), cone beam CT (CBCT) imaging and computer-assisted intraoperative navigation permits the assessment of the course of the mandibular canal throughout its length both prior to and during any surgical procedure that could place the inferior alveolar nerve and/or the mental nerve at risk ([Aizenbud et al 2012](#), [Wittwer 2012](#)).

Elongation of the coronoid process may be found bilaterally or unilaterally, resulting in progressive, painless restriction of mandibular opening, due to the impingement of the coronoid process on the medial aspect of the zygomatic arch(es). This rare condition is more common in males and usually presents in the middle of the third decade, although it has been reported in neonates. The aetiology is unknown. Treatment involves the resection of the coronoid process(es) ([Satoh et al 2006](#), [McLoughlin et al 1995](#), [Mulder et al 2012](#)).

Hyperplasia of the mandibular condyle is a rare unilateral condition that results in facial asymmetry and an altered occlusion (bite). The condition may occur at any age and, if it occurs prior to puberty, growth may not cease at the end of the growth period. Reports vary as to male:female ratio. Although condylar hyperplasia is said to be self-limiting, removal of the growth site in the condyle may be necessary to arrest the condition. Correction of the asymmetry of the jaw is usually performed; however, self-correction of the asymmetry has been reported. A variety of classification systems have been suggested ([Obwegeser and Make 1986](#), [Nitzan et al 2008](#), [Nitzan 2009](#), [Wolford 2014](#)).

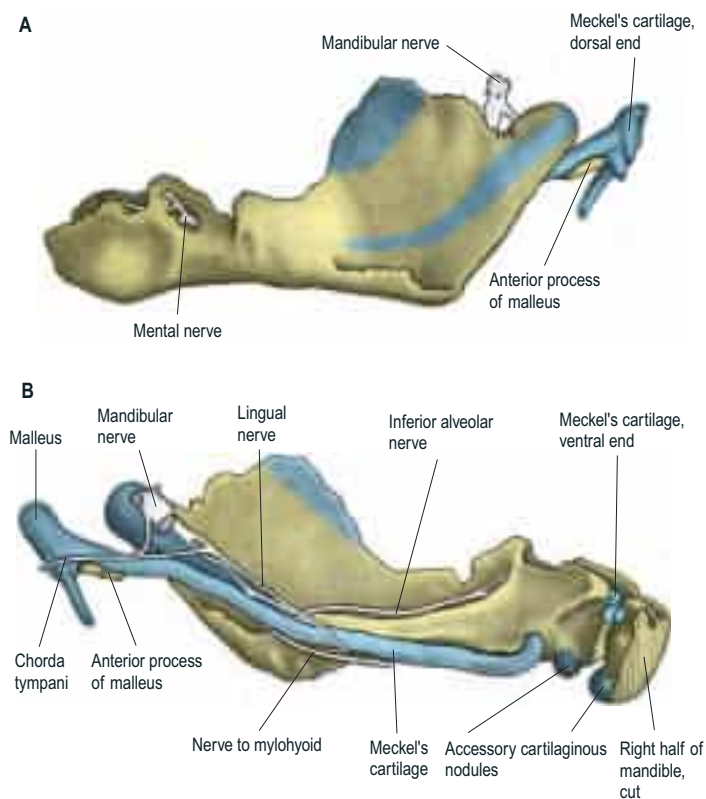


Fig. 32.6 The left half of a mandible of a human embryo, 95 mm long. **A**, Lateral aspect. **B**, Medial aspect. Blue: cartilage; yellow: bone. (Courtesy of A Low.)

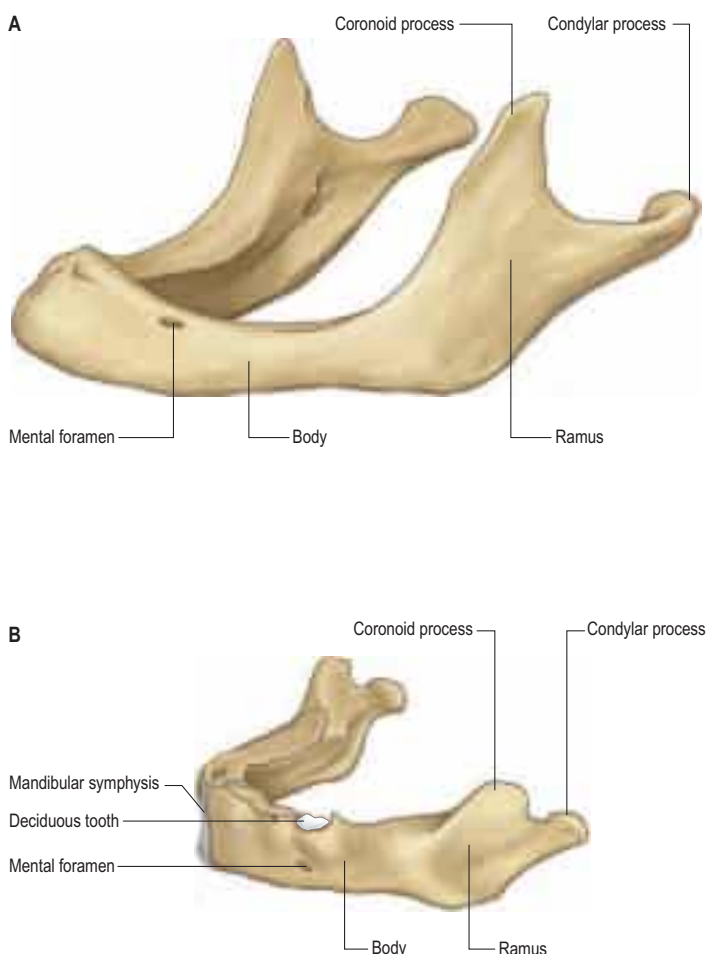


Fig. 32.7 **A**, An edentulous mandible; note the position of the mental foramen. **B**, A neonatal mandible. (Redrawn with permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

and ossification on its mandibular side proceeds towards the midline. When the latter process overtakes the former and ossification extends into median fibrous tissue, the symphysis fuses. At this stage, the body is a mere shell, which encloses the imperfectly separated sockets of deciduous teeth. The mandibular canal is near the lower border, and the mental foramen opens below the first deciduous molar and is directed forwards. The condyle is almost in line with the occlusal plane of the mandible and the coronoid projects above the condyle. During the first three postnatal years, the two halves join at their symphysis from below upwards, although separation near the alveolar margin may persist into the second year. The body elongates, especially behind the mental foramen, providing space for three additional teeth. During the first and second years, as a chin develops, the mental foramen alters direction; it no longer faces forwards but now faces backwards, as in the adult mandible, and accommodates the changing direction of the emerging mental nerve.

In general terms, increase in height of the body of the mandible is achieved primarily by formation of alveolar bone associated with the developing and erupting teeth, although some bone is also deposited on the lower border. Increase in length of the mandible is accomplished by deposition of bone on the posterior surface of the ramus and concomitant compensatory resorption on the anterior surface (accompanied by deposition of bone on the posterior surface of the coronoid process and resorption on the anterior surface of the condylar process); a part of the ramus is therefore modelled into an addition to the mandibular body. Increase in width of the mandible is produced by deposition of bone on the outer surface of the mandible and resorption on the inner surface. An increase in the comparative size of the ramus compared with the body of the mandible occurs during postnatal growth and tooth eruption.

The role of the condylar cartilages in mandibular growth remains controversial. One view states that continued proliferation of this cartilage is primarily responsible for the increase in both the mandibular length and the height of the ramus. Alternatively, there is persuasive experimental evidence that proliferation of the condylar cartilage is an adaptive response to function, rather than being genetically determined. Condylar growth and remodelling have been shown to be influenced significantly by local factors – notably, movement and loading of the temporomandibular joint – and to be relatively immune to systemic influences such as vitamin C and D deficiency. Considering the changes that occur in the dentition throughout life, continuous adaptation of the temporomandibular articulation is required in order to maintain functional occlusal alignment between the upper and lower arches of teeth; this adaptation is thought to be largely the result of ongoing condylar remodelling.

In adults, alveolar and subalveolar regions are about equal in depth, and the mental foramen appears midway between the upper and lower borders. If teeth are lost, alveolar bone is resorbed, and both the mental foramen and the mandibular canal come to lie closer to the superior border (see Fig. 32.7A). The mental foramen is placed higher than the mandibular canal posterior to it, and so resorption of the alveolus in edentulous patients exposes the nerve at the foramen, i.e. prior to the nerve in the mandibular canal (Cawood and Howell 1988, Lang 1995, Eufinger et al 1997).

Bone resorption in the mandible



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Blood supply of the mandible

The blood supply to the body of the mandible is derived from its principal nutrient artery, the inferior alveolar artery, and from the vessels supplying genioglossus, geniohyoid and the anterior belly of digastric (which are all attached to its lingual aspect between the mental foramina). The blood supply to these muscles is from the sublingual branch of the lingual artery and the submental branch of the facial artery. Branches of these vessels may perforate the lingual cortical plate through lateral lingual foramina in the premolar region and genial foramina in the midline, when these foramina are present (Fig. 32.10). Anastomoses between these vessels are common. A branch of the submental artery may anastomose with the mental artery, permitting retrograde vascular supply to the body and symphysis (relevant in mandibular fractures). The vascular supply to the mandibular symphysis is of importance in dental implant surgery. The ramus, including the mandibular angle, is supplied by the inferior alveolar artery and from the vessels supplying masseter and medial pterygoid. Vessels supplying temporalis supply the coronoid process. The vascular supply to the mandibular condyle, temporomandibular joint and the individual

The pattern of bone resorption (atrophy) of the mandible following the loss of teeth is said to be in a centrifugal ('down and out') direction, compared to a centripetal direction in the maxilla. These differences produce a progressive increase in the incongruity of the alveolar processes of the jaws. Bone loss is not limited to the alveolar part; it can involve the base (basal bone) to varying degrees. Progressive remodeling of the mandible and maxilla accompany the bone loss; it is not limited to the body of the mandible but also involves the coronoid process and the mandibular condyle, and reduction in both the height and width of the ramus. This bone loss has implications in denture construction, dental implant surgery and mandibular fractures (Figs 32.8–32.9) (Aziz and Najjar 2009, Ellis and Price 2008, Cawood and Howell 1988, Eufinger et al 1997, Sutton et al 2004).



Fig. 32.8 The opposing direction (arrows) of resorption ('atrophy') of the mandible and maxilla, resulting in increasing incongruity between the jaws. The resorption involves the alveolus and the basal bone. (Image courtesy of Dr Alf Nastri, Royal Melbourne Hospital.)

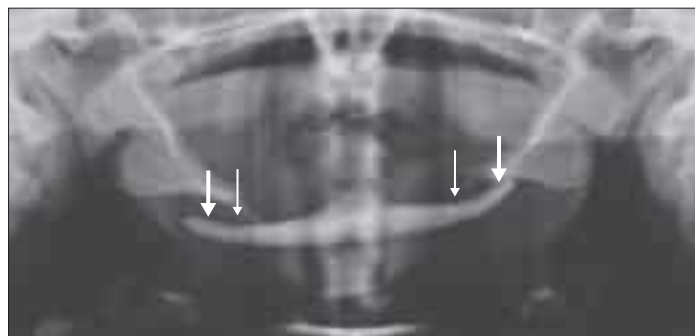


Fig. 32.9 A bilateral fracture of an edentulous atrophic mandible. There is complete loss of the alveolar process, extending into the underlying basal bone. The mental foramen (narrow arrows) and the mandibular canal (broad arrows) are at the superior border.

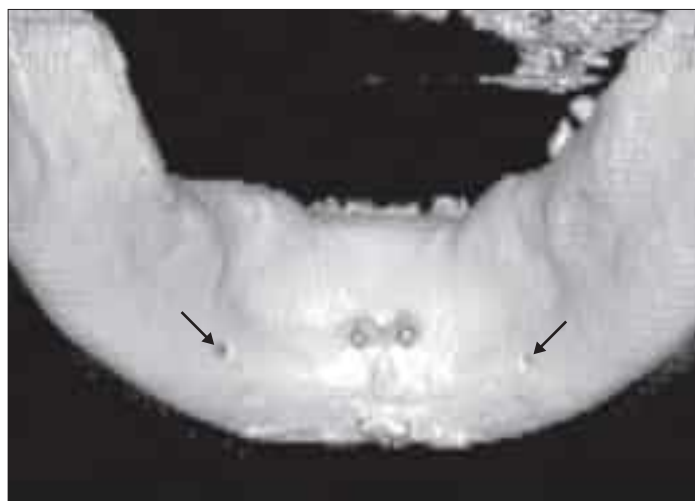


Fig. 32.10 A CT image showing the lateral lingual foramina (arrows). (Courtesy of Dr Koh Nakajima, with permission from Nakajima K, Tagaya A, Otonari-Yamamoto M, et al. Composition of the blood supply in the sublingual and submandibular spaces and its relationship to the lateral lingual foramen of the mandible. *Oral Surg Oral Med Oral Pathol Oral Radiol* 2014;117(1):e32–8.)

masticatory muscles is described below (Janfanza et al 2001, McDonnell et al 1994, Jacobs et al 2007, Romanos et al 2012, Nakajima et al 2014).

Endosseous blood supply to the mandible via lingual foramina



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TISSUE SPACES

Descriptions of the various 'spaces' in the head and neck and the fascial arrangements of the head and neck in the literature lack clarity, and are variable and open to misinterpretation. There is even controversy as to what constitutes 'fascia' (Guidera et al 2012). No gold standard exists as to what constitutes cervical fascia; however, the landmark paper by Grodinsky and Holyoke (1938) is suggested as a useful initial point of reference.

The contents of the infratemporal fossa are contained within a well-defined space that is bounded by and incorporates the masticatory muscles; this is termed the masticator space. The masticator space is closed posteriorly by the attachment of the deep cervical fascia, medial pterygoid and masseter (pterygomasseteric sling), and anteriorly, lateral to the ramus, by the firm attachment of masseter (the submasseteric space is a potential space). The masticator space is closed superiorly by both the temporal fascia laterally and the firm attachment of temporalis to the bone of the temporal fossa (the deep temporal space is also only a potential space). Medial to the ramus of the mandible, the pterygomandibular space (part of the masticator space) communicates freely with the pterygopalatine fossa, offering little resistance to the spread of infection or tumours. Posteriorly penetrating maxillary tumours directly involve the masticator space/infratemporal fossa and the pterygopalatine fossa. Resection of the masticator space has been termed a 'compartment' resection of the mandible (McMahon et al 2013, Blanchaert and Ord 1998) (Fig. 32.12).

Above the level of the zygomatic arch, the temporal fascia provides both the superior and lateral limits, by its attachments to the superior temporal line and the zygomatic arch, respectively. Below the zygomatic arch, the investing layer of deep cervical fascia (superficial layer of deep cervical fascia) splits into two laminae to enclose masseter, the lower part of temporalis and medial pterygoid, further defining the boundaries of the masticator space. The superficial (lateral) lamina covers masseter and attaches to the zygomatic arch, and the deep (medial) lamina runs on the deep surface of medial pterygoid, attaching to the skull base medial to the foramen ovale (Guidera et al 2014). This fascial covering of the masticatory muscles is felt to be distinct from the parotid fascia (capsule). The contents of the masticator space are temporalis, masseter, medial and lateral pterygoid, the ramus and coronoid process of the mandible, the mandibular nerve and otic ganglion, the maxillary artery and the pterygoid venous plexus, and part of the buccal fat pad anterolaterally. The submasseteric, pterygomandibular, superficial and deep temporal spaces are contained within.

The superficial temporal space lies between the temporal fascia laterally and temporalis medially. The deep temporal space, like the submasseteric space, is a potential space. It lies between temporalis laterally and the temporal fossa of the skull medially. Both spaces are in communication with the remainder of the masticator space inferiorly.

There is debate as to the upper limit of the masticator space, with the suggestion that it should be at the level of the zygomatic arch. Since both temporal spaces communicate with the remainder of the masticator space, a single masticator space incorporating the temporal fascia, muscles and temporal spaces is favoured (Guidera et al 2014).

TEMPOROMANDIBULAR JOINT

The temporomandibular joint (TMJ) is a synovial joint between the glenoid fossa (mandibular fossa) of the temporal bone above and the mandibular condyle. It is unusual in that its articular surfaces are lined by fibrocartilage, rather than hyaline cartilage; fibrocartilage is less susceptible to degeneration and has a greater repair capacity. The joint cavity is divided into upper and lower compartments ('spaces') by a fibrocartilaginous articular disc.

The anterior limit of the glenoid fossa is the articular eminence. This transverse ridge of dense bone, tilted down at an angle of approximately 25° to the occlusal plane, forms most of the articular surface of the mandibular fossa. It is strongly convex in the sagittal plane with a slight concavity in the coronal plane. It extends out laterally to the zygomatic

arch as the articular tubercle. The roof of the glenoid fossa, by contrast, is thin and often translucent when held to the light, confirmation that this is not a major load-bearing area of the joint (Hylander 2006). The posterior wall is formed by the tympanic plate, which also forms the anterior wall of the external acoustic meatus.

The anterior articular area of the fossa (articular fossa) is formed entirely from the squamous portion of the temporal bone and lined by articular tissue that extends anteriorly beyond the articular summit on to the preglenoid plane. The squamotympanic fissure marks the junction with the posterior non-articular (tympanic) area. The tegmen tympani, a bony plate of the petrous temporal, intervenes in the medial aspect of the fissure, where the squamotympanic fissure becomes the petrotympanic fissure. The postglenoid tubercle at the root of the zygomatic arch, just anterior to the squamotympanic fissure, separates the squamotympanic fissure laterally from the tympanic plate.

The condylar head, tilted forwards on the neck at an angle of approximately 30° (physiological anteversion), articulates with the fossa on its anterior and superior surfaces. Like the eminence, both its slope and shape are variable, influenced by age, function and dentition. In the coronal plane, its shape varies from that of a gable (particularly marked in those whose diet is hard) to roughly horizontal (in the edentulous).

It is probably impossible to measure the pressure developed on the articular surfaces of the human jaw joint when biting; however, direct measurement of loads across the joint in animals has demonstrated significant intermittent loading during mastication. There is also irrefutable theoretical evidence based on Newtonian mechanics that the jaw joint is a weight-bearing joint. With a vertical bite force of 500 N on the left first molar, the right condyle must support a load of well over 300 N (Osborn 1995). The non-working condyle is more loaded than the condyle on the working side, which may help explain why patients with a fractured condyle choose to bite on the side of the fracture.

FIBROUS CAPSULE

The lower part of the joint is surrounded by tight fibres, which attach the condyle of the mandible to the disc (collateral ligament). The upper part of the joint is surrounded by loose fibres, which attach the disc to the temporal bone (Fig. 32.13). Thus the articular disc is attached separately to the temporal bone and to the mandibular condyle, forming what could be considered two joint capsules. These attachments stabilize the disc but allow rotation over the condyle. Longer fibres joining the condyle directly to the temporal bone may be regarded as reinforcing. The capsule is attached above to the anterior edge of the preglenoid plane, posteriorly to the lips of the squamotympanic fissure, between these to the edges of the articular fossa, and below to the periphery of the neck of the mandible.

LIGAMENTS

The ligaments of the temporomandibular joint are the temporomandibular (lateral) ligament, the sphenomandibular ligament and the stylomandibular ligament.

Temporomandibular (lateral) ligament

The broad temporomandibular ligament, reinforcing the joint capsule laterally, is attached above to the articular tubercle on the root of the zygomatic process of the temporal bone (see Fig. 32.13). It extends downwards and backwards at an angle of approximately 45° to the horizontal, to attach to the lateral surface and posterior border of the neck of the condyle, deep to the parotid gland. A short, almost horizontal, band of collagen connects the articular tubercle in front to the lateral pole of the condyle behind. It may function to prevent posterior displacement of the resting condyle and also initiates translation of the condyle on mouth opening.

Sphenomandibular ligament

The sphenomandibular ligament is medial to, and normally separate from, the capsule (see Fig. 32.13). It is a flat, thin band that descends from the spine of the sphenoid and widens as it reaches the lingula of the mandibular foramen; it has an average width at its insertion into the mandible of about 12 mm. Some fibres traverse the medial end of the petrotympanic fissure and attach to the anterior malleolar process. This part is a vestige of the dorsal end of Meckel's cartilage.

The endosseous supply of the mandibular symphysis via the lingual foramina, and the rich vascular supply to the anterior floor of the mouth are of particular importance in dental implant surgery and osteotomies, e.g. genioplasty, at this site.

Foramina at the level of, or above, the genial tubercle (superior genial foramen) and below the genial tubercle (inferior genial foramen) are almost always found (McDonnell et al 1994, Jacobs et al 2007). Lateral lingual foramina are usually identified, with an average distance above the lower border of the mandible of 5 mm (Romanos et al 2012). The superior genial foramen may contain a branch of the lingual artery and nerve. The inferior foramen may contain a branch of the sublingual or submental arteries and a branch of the mylohyoid nerve. The lateral foramen may contain a branch of the submental artery (Jacobs et al 2007, Nakajima et al 2014). Although the relative contributions from the individual vessels vary, anastomoses between these perforating vessels and the inferior alveolar artery, within the mandible, are common. The soft tissues in the anterior floor of the mouth adjacent to the mandible are supplied by contributions from the sublingual, submental and mylohyoid arteries; there are considerable anastomoses between them (Romanos et al 2012, Nakajima et al 2014, Loukas et al 2008, Flanagan 2003). Brisk, potentially life-threatening haemorrhage has been reported with dental implant placement in the mandibular symphysis, usually associated with perforation of the lingual cortical plate of the mandible (Romanos et al 2012, Loukas et al 2008, Woo et al 2006, Dubois et al 2010) (Fig. 32.11) and death has resulted from genioplasty, secondary to haemorrhage in the anterior floor of the mouth (personal communication). The precise source of the haemorrhage can be difficult to detect and may arise from both within the mandible and the soft tissues in the floor of the mouth, from any of the above-named vessels. The anastomoses between the vessels may result in haemorrhage on both sides of mylohyoid and thus into the sublingual and the submental/submandibular spaces bilaterally. As the lateral and midline lingual foramina are closer to the lower border of the mandible, caution with longer implants has been recommended in this region.



Fig. 32.11 A three-dimensional reformatted CT image showing an implant perforating the lingual cortical plate (arrow), leading to haemorrhage in the anterior floor of the mouth. A tracheostomy was required to secure the airway. (With permission from Dubois L, de Lange J, Baas E, Van Ingen J. Excessive bleeding in the floor of the mouth after endosseous implant placement: a report of two cases. *Int J Oral Maxillofac Surg* 2010;39:412–415.)

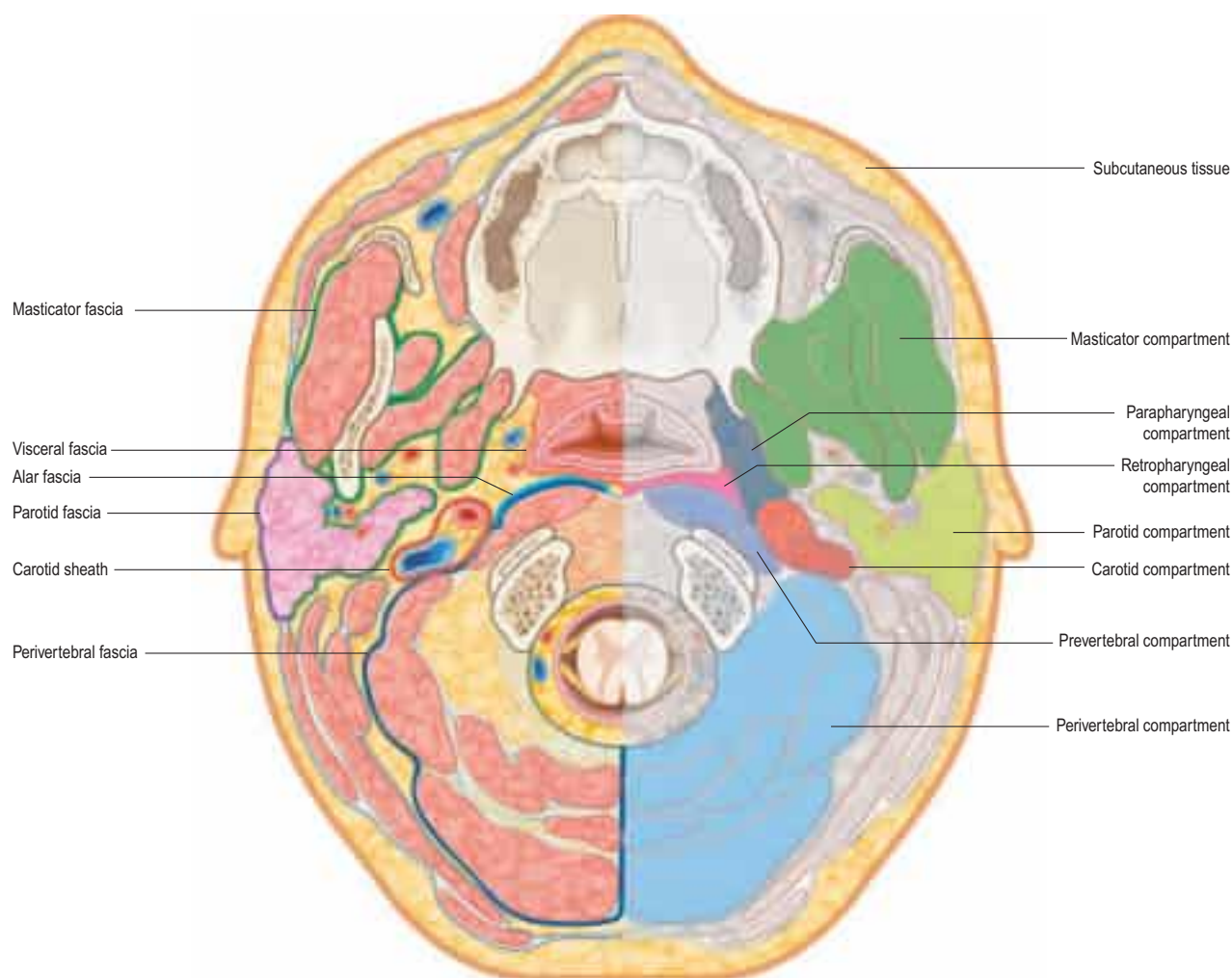


Fig. 32.12 A transverse section at the level of the hard palate demonstrating the cranial fascia (left) and associated compartments (right). (With permission from Guidera, A. K., Dawes, P. J. D., Fong, A., Stringer, M. D. (2014), *Head and neck fascia and compartments: No space for spaces*. *Head Neck*, 36:1058–1068.)

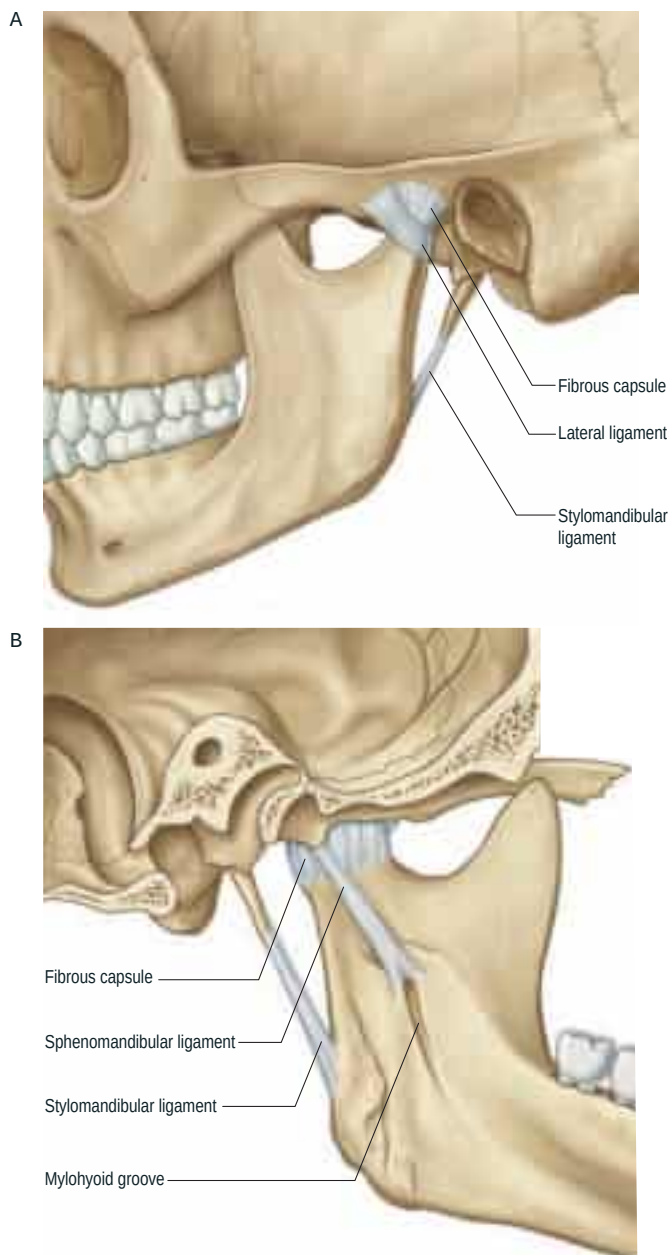


Fig. 32.13 The left temporomandibular joint. **A**, Lateral aspect. **B**, Medial aspect.

With the jaw closed, there is approximately 5 mm slack within the ligament, but it becomes taut when the jaw is about half open. Superiorly, lateral pterygoid, the auriculotemporal nerve and the maxillary artery and vein are lateral to the ligament, the nerve and vessels running between the ligament and the condylar neck. The chorda tympani lies medially and occasionally grooves the spine of the sphenoid. Antero-inferiorly, the inferior alveolar nerve and vessels and a parotid lobule separate the ligament from the ramus of the mandible. The vessels and nerves to mylohyoid pierce the ligament adjacent to the lingula. Medial pterygoid is inferolateral. The sphenomandibular ligament is separated from the pharynx by fat and a pharyngeal vein.

Stylomandibular ligament

The stylomandibular ligament is a thickened band of deep cervical fascia that stretches from the apex and adjacent anterior aspect of the styloid process to the angle and posterior border of the mandible (see Fig. 32.13). Its position and orientation indicate that it cannot mechanically constrain any normal movements of the mandible and does not seem to warrant the status of a ligament of the joint.

SYNOVIAL MEMBRANE

The synovial membrane lines the inside of the capsule of the joint but does not cover the disc or the articular surfaces (condyle, fossa or

articular eminence). The synovium is most abundant in the bilaminar zone of the articular disc, forming loose folds posteriorly when the condyle is positioned in the glenoid fossa. These folds disappear when the condyle is protruded and the synovium is stretched.

Synovial fluid and joint lubrication

The coefficient of friction between the articulating surfaces of the temporomandibular joint is almost zero (Nickel et al 2006), reflecting the net result of a combination of adequate joint lubrication, the surface structure of the articulating surfaces and the articular disc. Joint lubrication depends on the synovial fluid, which also provides protection and nutrition to the articular surfaces and is the sole supplier of nutrients to the avascular disc. Surface active phospholipids (SAPLs) and hyaluronic acid are the key lubricants of the joint and act to protect the articular surfaces. It has been hypothesized that abnormalities of the joint lubrication system, e.g. changes resulting from uncontrolled oxidative stress, may play a role in the onset of temporomandibular joint disorders (Nitzan 2001).

ARTICULAR SURFACES

The anterior and superior surfaces of the mandibular condyle, and the articular eminence and pregenoid plane of the squamous temporal bone are the principal articulating surfaces of the joint. This is reflected in the thickness of the articular tissue covering these surfaces. The fibrocartilaginous covering of the condyle is composed of four distinct layers (De Bont et al 1984). The most superficial layer consists of densely packed fibres of type I collagen that is arranged mostly parallel to the articular surface and aligned in an anteroposterior direction (this can be seen as striation on arthroscopy – Video 32.1). This covers a thin cellular layer, the proliferative zone that is continuous with the cambial layer of the periosteum beyond the margins of the joint. The third layer, of hypertrophic cartilage, is rich in intercellular matrix; it contains chondrocytes scattered throughout its depth, and randomly orientated fibres of collagen type II. The fourth layer, immediately above the subchondral bone, is the zone of calcification. Although the number of chondrocytes within the hypertrophic zone decreases with age, undifferentiated mesenchymal cells have been identified in postmortem specimens of all ages (Hansson et al 1977). This indicates that a capacity for proliferation and repair persists in condylar cartilage, and may be the reason why condylar remodelling occurs throughout life (Robinson 1993, Toller 1974).

ARTICULAR DISC

The articular disc divides the joint into a superior (discotemporal) space and an inferior (discomandibular) space, both filled with synovial fluid. The transversely oval disc is composed of avascular dense fibrous connective tissue with some chondrification in areas of maximum loading (Fig. 32.14). It has a thick margin, which forms a peripheral anulus and a central depression in its lower surface that accommodates the articular surface of the mandibular condyle. The depression probably develops as a mechanical response to pressure from the condyle as it rotates inside the anulus. The disc is stabilized on the condyle in three ways. Its edges are fused with the part of the capsular ligament that tightly surrounds the lower joint compartment and is attached around the neck of the condyle; well-defined bands in the capsular ligament attach the disc to the medial and lateral poles of the condyle (collateral ligaments), and additionally, the thick anulus prevents the disc sliding off the condyle (self-centring capability – see below), provided that the condyle and disc are firmly lodged against the articular fossa (as is normally the case).

In sagittal section, the disc has three distinct parts: an anterior band, a thinner intermediate zone and a posterior band. The posterior band is the thickest. Its upper surface is concavo-convex where it fits against the convex articular eminence and the concavity of the articular fossa. Posteriorly, the disc is attached to a richly vascularized and innervated region, the bilaminar zone (posterior attachment), which splits into two laminae; unlike the rest of the disc, its normal function is to provide attachment rather than intra-articular support. The upper lamina is composed of fibroelastic tissue and is attached to the squamotympanic fissure. There is a condensation of elastic fibres in the medial portion of the upper lamina, which is visible as the posterior oblique protuberance on arthroscopy (see Video 32.1). The lower lamina is composed of fibrous non-elastic tissue and is attached to the posterior aspect of the condyle. The intervening connective tissue between the upper and lower laminae contains a vascular plexus and is well innervated, but the

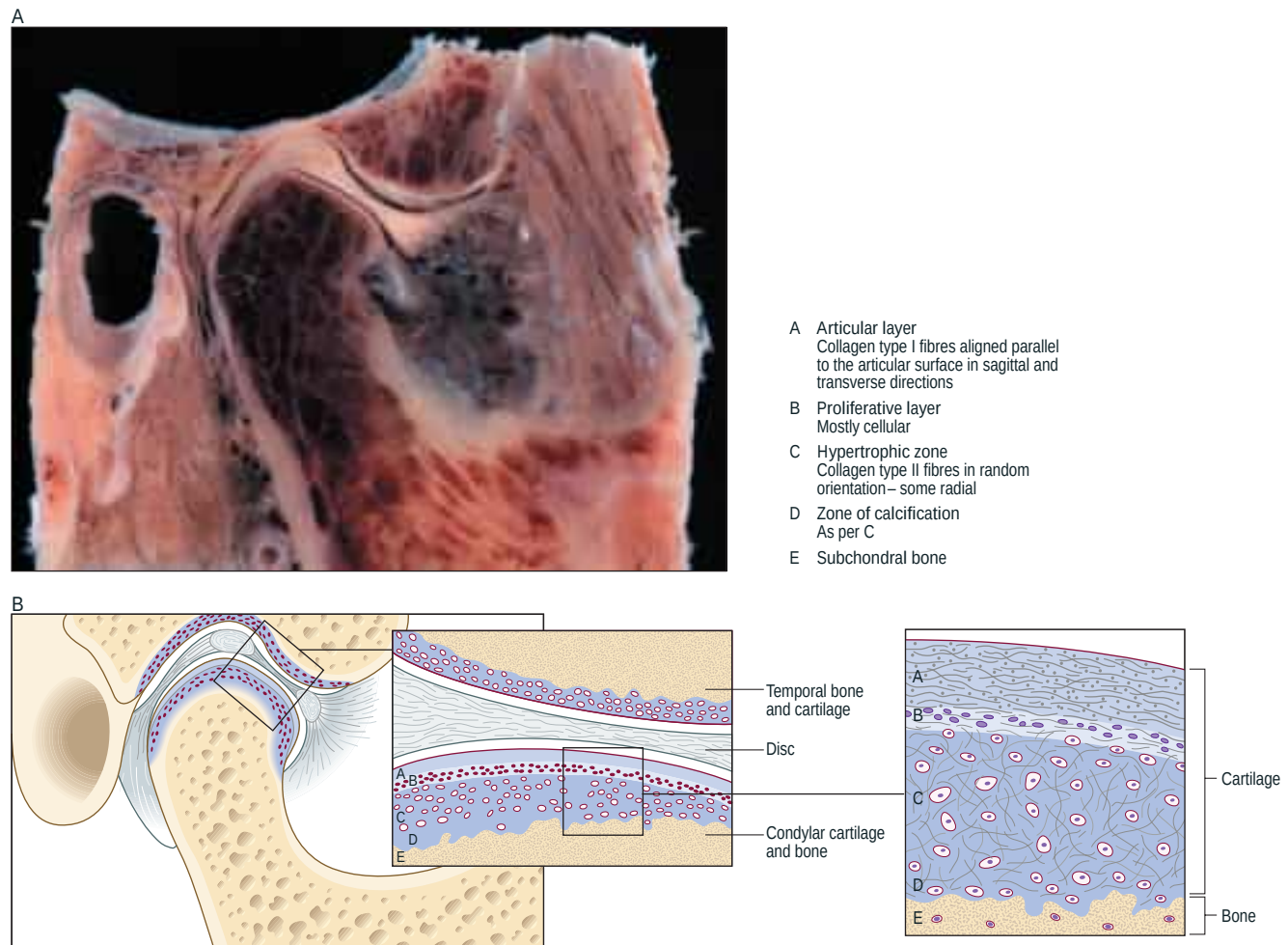


Fig. 32.14 Sagittal sections through the right temporomandibular joint. **A**, Note the relationship of the external acoustic meatus and articular disc to the mandibular condyle. **B**, The microstructure of condylar cartilage. (A, Courtesy of Professor Cristina Manzaneres Céspedes, Anatomy and Human Embryology Unit, Bellvitge Campus, University of Barcelona.)

central part of the disc itself is avascular and not innervated. With anterior movement of the disc–condyle complex, the volume of the posterior attachment increases as a result of the negative pressures generated, resulting in an influx of blood into the venous plexus. This pumping mechanism is important in the nutrition of the joint (Stegenga and de Bont 2006).

Upper and lower joint compartments

The upper joint space contains approximately 1.2 ml of synovial fluid and the lower approximately 0.9 ml. Both spaces may be considerably distended with fluid for diagnostic purposes in temporomandibular joint arthrography, for therapeutic purposes in arthrocentesis, and to improve access and visibility in arthroscopy. Both joint spaces have an anterior and posterior recess or pouch, and are lined by synovium, which has a grey hue when viewed through an arthroscope (see Video 32.1).



Functions of the articular disc

The articulating surfaces of the mandibular condyle and the articular fossa are highly incongruent. Wear of the articular surfaces may be reduced by the presence of a deformable articular disc, which not only distributes compressive joint loading over a larger contact area but also significantly reduces the frictional force on the condyle and the articular fossa by separating slide and rotation into different joint compartments.

The disc may aid joint lubrication by storing fluid squeezed out from loaded areas to create a weeping lubricant. In addition, its flexibility, due to its high water content and viscoelastic properties, allows it to function almost like a fluid, further reducing friction within the joint and thereby aiding joint lubrication. The friction between the articulating surfaces in the absence of the disc is 2–3 times greater than with the disc *in situ*.

The role of the disc in relation to joint stability is controversial. One view is that a deformable and viscoelastic disc stabilizes the condyle within the joint by filling the changing and irregular joint space created

during movement. The disc moves in an almost congruent manner relative to the condyle controlled by neuromuscular forces: the upper head of lateral pterygoid anteriorly and the elastic tissue in the bilaminar region posteriorly together pull the disc forwards or backwards. In addition, the shape of the disc with its thick, 'self-seating' anulus helps to prevent the disc sliding off the condyle. An alternative view is that a slippery articular disc doubles the number of virtually friction-free sliding surfaces, and so destabilizes the condyle in the same way that stepping on a banana skin destabilizes the foot. These two roles are not mutually exclusive. A deformable disc stabilizes the condyle within the joint whilst, by doubling the friction-free surfaces, the disc–condyle assembly slides freely under compressive loading and so is 'destabilized'.

RELATIONS

Superiorly, the thin roof of the glenoid fossa has a mean thickness of 0.9 mm (range 0.2–4 mm) and separates the upper joint space from the middle cranial fossa. This bony partition is occasionally fenestrated by inflammatory processes or inadvertently during surgery to the joint, particularly for release of ankylosis, and rarely by the condyle being driven superiorly by violent trauma to the mandible. The maxillary artery and its proximal branches, most notably the middle meningeal artery, lie medially, just beyond the joint capsule. The uppermost part of the parotid capsule enclosing the branches of the facial nerve that supply the muscles of the upper face, including orbicularis oculi, are lateral to the joint capsule; the thread-like nerves are at risk during surgical approach to the joint. The upper part of the infratemporal fossa, which contains the two heads of lateral pterygoid, is anterior to the condyle (Fig. 32.15). It is into this space anterior to the articular eminence that the condyle is displaced in cases of dislocation of the temporomandibular joint or fracture dislocation of the condylar head due to the vector of pull from lateral pterygoid, which inserts into the anterior condylar neck. Posteriorly, the tympanic plate and tegmen tympani separate the joint from the middle ear cavity. Perforation through this

The articular disc is designed to withstand compressive and shearing loads, and has viscoelastic properties, allowing it both to deform when loaded and to conform to the surfaces causing this deformation. Removal of the disc increases surface friction within the joint by 2–3 times compared to joints with an intact disc. Long-term follow-up of patients who have had the disc removed ('discectomy') demonstrates that the joint undergoes significant remodelling similar to that seen in osteoarthritis. There is flattening, slight roughening and sclerosis of the condylar head and articular fossa. These changes, which are believed to be adaptive responses of the condyle and fossa to the absence of an intervening disc, collectively produce a broader contact, decreasing the load per unit area.

The main purpose of collagen in tissue is to resist tensile forces. The orientation of the collagen fibres in the disc varies in the individual regions, reflecting the direction of the tensile forces imparted to the disc. In the intermediate zone, they are predominantly orientated in an anteroposterior direction. Around the periphery, the collagen forms a ring-like structure with a three-dimensional framework of collagen fibrils. In the posterior and anterior bands, the predominant orientation is lateral–medial and superior–inferior. Type 1 collagen, similar to that found in the superficial layers of the head of the condyle and the articular fossa, predominates in the disc.

Chondroitin sulphate and dermatan sulphate are the principal glycosaminoglycans (GAGs) in the intercellular matrix. GAGs are very hydrophilic and, as such, play an important role in the resistance and distribution of compressive load applied to the temporomandibular joint. Their concentration varies with the area of the disc, being highest in the superficial layer of the intermediate zone ([Nickel and McLachlan 1994](#), [Tanaka et al 2006](#), [Milorio and Henrickson 2010](#), [Detamore and Athanasiou 2003](#), [Obrez and Gallo 2006](#), [Paegle et al 2003](#)).

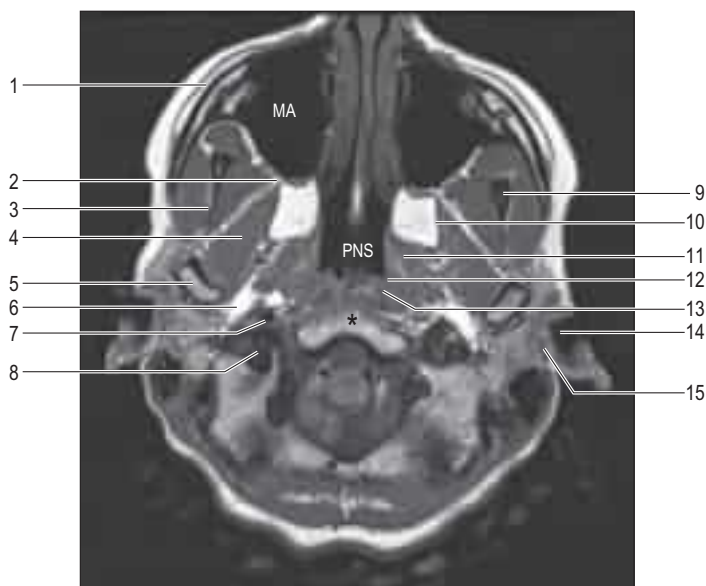


Fig. 32.15 An axial T1 weighted MRI at the level of the external acoustic meatus. Key: 1, body of zygoma; 2, pterygopalatine fossa; 3, temporalis; 4, lateral pterygoid; 5, condyle; 6, parapharyngeal fat stripe; 7, internal carotid artery; 8, jugular bulb/foramen; 9, coronoid process; 10, lateral pterygoid plate; 11, torus tubarius; 12, fossa of Rosenmüller; 13, longus colli; 14, external auditory meatus; 15, superficial lobe of parotid gland. Abbreviations: MA, maxillary antrum; PNS, postnasal space. *, anterior arch of atlas.

thin, bony wall into the middle ear is a recognized complication of temporomandibular joint arthroscopy.

VASCULAR SUPPLY AND INNERVATION

The articular tissues and the dense part of the articular disc have no nerve supply. The nerve supply to the temporomandibular joint is from branches from the mandibular division of the trigeminal nerve, mostly through the auriculotemporal branch, along with branches from the masseteric and deep temporal nerves. Postganglionic sympathetic nerves supply the tissues associated with the capsular ligament and the looser posterior bilaminar extension of the disc. The temporomandibular joint capsule, lateral ligament and retroarticular tissue contain mechanoreceptors and nociceptors. The input from mechanoreceptors provides a source of proprioceptive sensation that helps control mandibular posture and movement.

The joint derives its arterial supply from the superficial temporal artery laterally and the maxillary artery medially. Penetrating vessels that supply lateral pterygoid may also supply the condyle. Veins drain the anterior aspect of the joint and associated tissues into the plexus surrounding lateral pterygoid; posteriorly, they drain into the vascular region that separates the two laminae of the bilaminar region of the disc. Positive and negative pressure produced by forward and backward movement of the condyle shunts blood between these regions. Lymphatics drain deeply to the upper cervical lymph nodes surrounding the internal jugular vein.

JAW MOVEMENTS

Movements of the condyle in the temporomandibular joint

The major function of the mandible is to exert, via the teeth, the force necessary to break down food into smaller particles and so facilitate digestion. Pure vertical movements of the lower teeth create a crushing force that is ineffective in breaking up tough fibrous food. Humans use a lateral movement of the lower jaw to create a shear force that enhances the effectiveness of the power stroke of mastication. Bodily lateral movement of the whole jaw, the Bennett shift, is insignificant. Extensive lateral movement is only possible when the jaw is rotated horizontally about one condyle while the other condyle slides backwards and forwards. The temporomandibular joint is structurally adapted to accommodate both sliding/translation and rotation/hinging in a sagittal plane. Sliding/translation occurs because the capsular ligament that surrounds the upper joint compartment is loose, whereas the capsular

ligament that encloses the lower joint compartment is tight, and only allows the condyle to rotate over the depression inside the anulus of the articular disc. The normal range of maximum mouth opening measured between the upper and lower incisor tips is 35–50 mm, the first 15–25 mm of which is due to hinging of the condyle; the remainder is due to condylar translation. The adult range of movements is reached at around 10 years in females and 15 years in males. Lateral excursions of the chin can normally achieve 8–12 mm of movement at the lower incisors. At the limit of normal opening, the condyle has translated on to the peak or just beyond the peak of the articular eminence. Translation beyond this results in subluxation and dislocation of the joint.

Symmetrical opening

Symmetrical jaw opening is associated with preparation for incising. At the start, each mandibular condyle rotates in the lower joint compartment inside the anulus of its disc. After a few degrees of opening, the condyle continues rotating inside its disc and, in addition, both slide forwards down the articular eminence of the upper joint compartment. Without this forward slide or translation, it becomes impossible to continue opening the jaw beyond a gape of approximately 25 mm.

Opinions differ as to why this forward slide should occur; the fact that the condyle in a cadaver still slides forwards when the jaw is rotated open suggests that it is not a neuromuscular response but rather the result of mechanical constraints. When the jaw is rotated open, the temporomandibular ligament rapidly becomes taut (*Osborn 1995*). The taut ligament acts as a constraint that allows the mandible only two rotary movements: it can swing about the upper attachment of the ligament and rotate about the lower attachment. The lower end of the taut ligament acts as a moving fulcrum that converts the downward and backward pull of the opening rotary force (created at the front by digastric and geniohyoid) into one that drives the condyle upwards and forwards into the concavity of the overlying articular disc. This now pushes the disc forward. Swing about the upper attachment creates space above for the disc to slide further forwards, which is possible because the upper part of the capsular ligament is loose. The two movements, rotation and swing, are inextricably linked by the taut ligament and, via the condyle, combine to keep the disc in firm contact with the articular eminence while the jaw is opened. The disc is stabilized by its tight attachment to the condyle (collateral ligament) and by the thickened margins of its anulus that prevent it sliding through the thinner compressed region between the centre of the condyle and the articular eminence.

As forward slide of the condyle continues, the controlling influence exerted by the temporomandibular ligament diminishes. The lingula of the mandible moves away from the spine of the sphenoid, tautening the originally slack sphenomandibular ligament, which now acts in the same way as the temporomandibular ligament, to maintain the condyle against the articular eminence. Symmetrical opening thus appears to consist of at least three separate phases: an early phase controlled by the temporomandibular ligament and articular eminence; a short middle phase in which either both temporomandibular and sphenomandibular ligaments, or neither, act to constrain movements; and a late phase controlled by the sphenomandibular ligament and articular eminence.

Changes in disc position during movement

With the teeth in occlusion, the condyle is in the glenoid fossa and the intra-articular disc sits on the condylar head, with its posterior band at the 12 o'clock position from the centre of the condylar head in the sagittal plane (*Fig. 32.16A*). As mouth opening begins (*Fig. 32.16B*), the condyle rotates within the lower joint space and the disc remains stationary. At about mid-opening, the condyle and disc begin to move forwards together so that their relative position is maintained (*Fig. 32.16C*). At maximum gape (*Fig. 32.16D*), the condylar head slides further anteriorly than the disc; the disc is rotated backwards relative to the condyle by the pull of the elastic superior lamina of the posterior attachment, so that the anterior band of the articular disc is above the summit of the condyle. This differential movement results in the disc moving only about half the distance of the condyle. Mouth closure involves the disc moving back in tandem with the condyle under the influence of the elastic fibres in the upper lamina of the bilaminar area posterior to the disc itself (*Fig. 32.16E*), which also reverses its angular rotation (*Fig. 32.16F*) to reassume the starting position of the cycle.

The changes in disc position may become out of phase with the condyle, causing obstruction to smooth mandibular movement. The disc is typically pulled anteromedially by the lateral pterygoid, so preventing forward translation of the condyle. With the disc in this position, the posterior band of the disc is anterior to the condyle, which

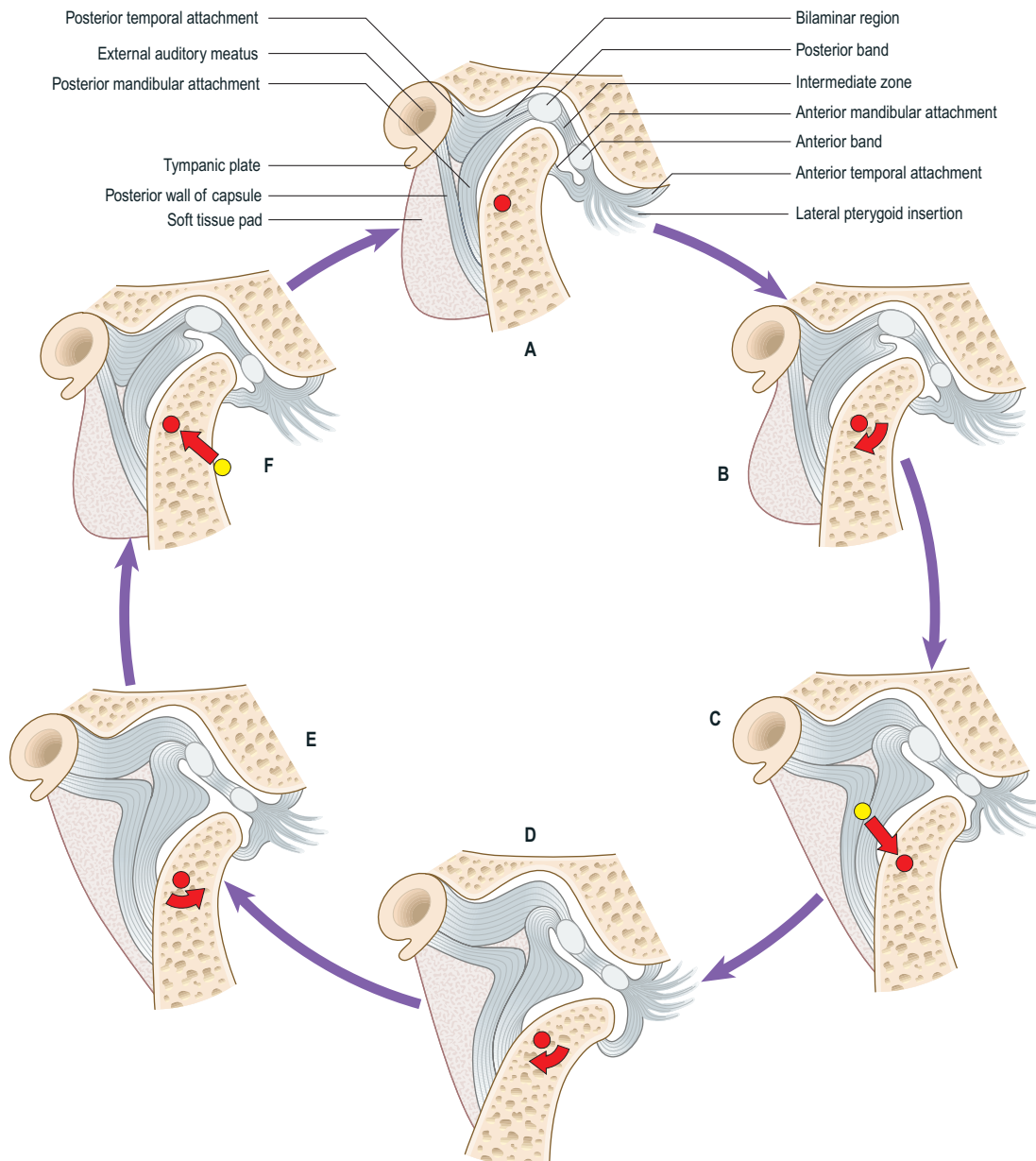


Fig. 32.16 Changing relationships of the condyle of the mandible, the articular disc and the articular surface of the temporal bone during one complete opening (A → D) and closing (D → A) cycle of the mouth.

causes an obstruction to forward translation of the condyle. As the force from the condyle on the disc increases, the elastic posterior attachments of the disc are stretched until the resistance to movement is overcome and the disc reduces into a normal relationship with the condyle, which is associated with an audible click (audible to the patient and occasionally to others). Once the normal disc–condyle relation is re-established, the mouth is free to achieve maximum opening (see Fig. 32.16C,D). The disc may then snap back in front of the condylar head and does not reduce into a normal relationship with the condyle during opening, the condyle is prevented from further forward translation and mouth opening is restricted. This is a common clinical condition and is known appropriately as closed lock or disc displacement without reduction (see below).

Eccentric jaw opening

Jaw movement during mastication can be divided into three parts: opening, closing and power strokes. Eccentric jaw opening is associated with preparing for the power stroke of mastication. The mandibular condyle on the non-working side slides back and forth during lateral movements associated with the power stroke on the working side. Although the jaw muscles now have the major control over mandibular movements, the temporomandibular and sphenomandibular ligaments keep the condyle firmly against its articular eminence during opening.

Eccentric and symmetrical jaw closing

During closure of the jaw, the masticatory muscles act in combination to force the joint surfaces together. This compresses the joint tissues and

potentially shortens the ligaments so that they no longer constrain jaw movements. Under these conditions, jaw movements and the positions of the condyles are controlled by neuromuscular processes (within the limits of constraints imposed by the articular eminence, the occluding surfaces of the teeth and the presence of food between them). Note that the non-working condyle moves the furthest, and is the most heavily loaded, during the power stroke of mastication. The loads on each joint, balancing and working, drive each condyle more forcefully into its articular eminence.

The envelope of motion

The envelope of motion is the volume of space within which all movements of a point on the mandible have to occur because the limits are set by anatomical features, i.e. by the shape or size of the upper and lower jaws, by tooth contacts, and by the attachment of muscle and ligaments. The mandibular movements at these extremes are termed 'border movements'.

In consciously controlled movement of the jaw from the rest position to the fully opened position, the trajectory of the mandibular incisal edge is two-phased. The first phase is a hinge-like movement during which the condyles are retruded within the mandibular fossae. When the teeth are opened by approximately 25 mm, the second phase of opening occurs by anterior movement or protrusion of the condyles down the articular eminences with further rotation.

If conscious effort is used, a closure path can then be followed in which the jaw is closed to an extreme protruded tooth contact position, after which it has to be retruded to the starting position. Similar considerations apply to lateral movements; mandibular rotation around a retruding condyle and the protraction of the opposite condyle are

anatomically limiting factors that are again rarely encountered in normal function.

Although all mandibular movement occurs within or along the border movements, knowledge of the extent of border movements provides little information on the nature of the movements during chewing or on the free ('empty'), habitual movements during both opening and closing, which have a more limited trajectory.

Disc position and internal derangement

Considerable attention has been directed at abnormalities of disc position (collectively known as internal derangements) as a cause of joint noises (clicking), limitation of jaw movement and/or pain. The position of the disc is considered to be normal when, with the mandible at rest, the posterior band is positioned in the 12 o'clock position from the centre of the condyle in the sagittal plane. Displacement of the disc anteromedially is most common. MRI and arthrography have demonstrated that, although disc position and clinical symptoms are often linked, a significant proportion of subjects with apparently normal temporomandibular joint function have disc displacement on imaging and, conversely, patients with normally positioned discs may have severe restriction of mouth opening. Similarly, individuals with displaced discs who have undergone disc repositioning surgery often show functional improvement despite persistent displacement of their intra-articular disc and vice versa.

It is appreciated that limited mouth opening is not always related to disc displacement or non-reducibility but may be the result of impaired joint lubrication and disc mobility. It is also suggested that, in an otherwise asymptomatic jaw with normal movements, an anteriorly 'displaced' disc could be considered a normal variant (Stegenga and de Bont 2006) or, at the least, not an indication for treatment.

It would appear that the adaptive remodelling capacity of the joint (both the mandibular condyle and articular fossa) and adequate joint lubrication are important for maintaining function of the temporomandibular joint.

DEVELOPMENT OF THE TEMPOROMANDIBULAR JOINT

The bony components of the secondary jaw joint develop in membrane as two separate condensations of neural crest mesenchyme between the eighth and twelfth week *in utero*. This arrangement is different from that seen with most synovial joints, where a single blastema cavitates to form a joint space and the articulating bones are derived from the same condensation of tissue. The separate development of the temporal and condylar parts of the temporomandibular joint means that the articular surface of each bone is covered by a persisting layer of proliferative cells that is continuous with the periosteum surrounding each separate bony component (Fig. 32.17).

MUSCLES

The four principal muscles of mastication are medial and lateral pterygoid, temporalis and masseter; their actions produce movements of the mandible at the temporomandibular joints.

The infratemporal fossa contains medial and lateral pterygoid and the tendon of temporalis. Masseter lies on the face, on the lateral surface of the ramus of the mandible, but will be considered here.

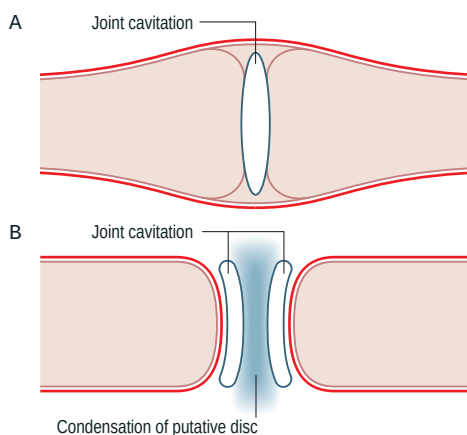


Fig. 32.17 A,B, Development of the temporomandibular joint; the temporal and condylar components of the joint develop from two separate condensations of neural crest mesenchyme.

Masseter

Masseter (Fig. 32.18) is a quadrilateral muscle that consists of three layers that blend anteriorly. The superficial layer is the largest. It arises by a thick aponeurosis from the maxillary process of the zygomatic bone and from the anterior two-thirds of the inferior border of the zygomatic arch. Its fibres pass downwards and backwards, to insert into the angle and lower posterior half of the lateral surface of the mandibular ramus. The superficial fibres are angled approximately 10° from the vertical, as is visible in lean individuals. In the coronal (frontal) plane, the muscle forms a 10° angle with the mandibular ramus. Intramuscular tendinous septa in this layer are responsible for the ridges on the surface of the ramus. The middle layer of masseter arises from the medial aspect of the anterior two-thirds of the zygomatic arch and from the lower border of the posterior third of this arch. It inserts into the central part of the ramus of the mandible. The deep layer arises from the deep surface of the zygomatic arch and inserts into the upper part of the mandibular ramus and into its coronoid process. The deep fibres run vertically, and are evident just anterior to the temporomandibular joint, where they are not covered by the more superficial layers. There is still debate as to whether fibres of masseter are attached to the anterolateral part of the articular disc of the temporomandibular joint. Hypertrophy of the masseter may occur either unilaterally or bilaterally. The aetiology is uncertain and there may be associated prominence of the mandibular angle. Treatment may involve surgical reduction of the deep aspect of the muscle and contouring of the mandibular angle or injections of botulinum toxin to paralyse the motor nerves.

Relations Skin, platysma, risorius, zygomaticus major, the parotid gland and duct, branches of the facial nerve and the transverse facial branches of the superficial temporal vessels are all superficial relations. Temporalis and the ramus of the mandible lie deep to masseter. The anterior margin of masseter is separated from buccinator and the buccal branch of the mandibular nerve by a buccal pad of fat and crossed by the facial vein. The posterior margin of the muscle is overlapped by the parotid gland. The masseteric nerve and artery reach the deep surface of masseter by passing over the mandibular incisure (mandibular notch).

Vascular supply Masseter is supplied by the masseteric branch of the maxillary artery, the facial artery and the transverse facial branch of the superficial temporal artery.

Innervation Masseter is supplied by the masseteric branch of the anterior trunk of the mandibular nerve.

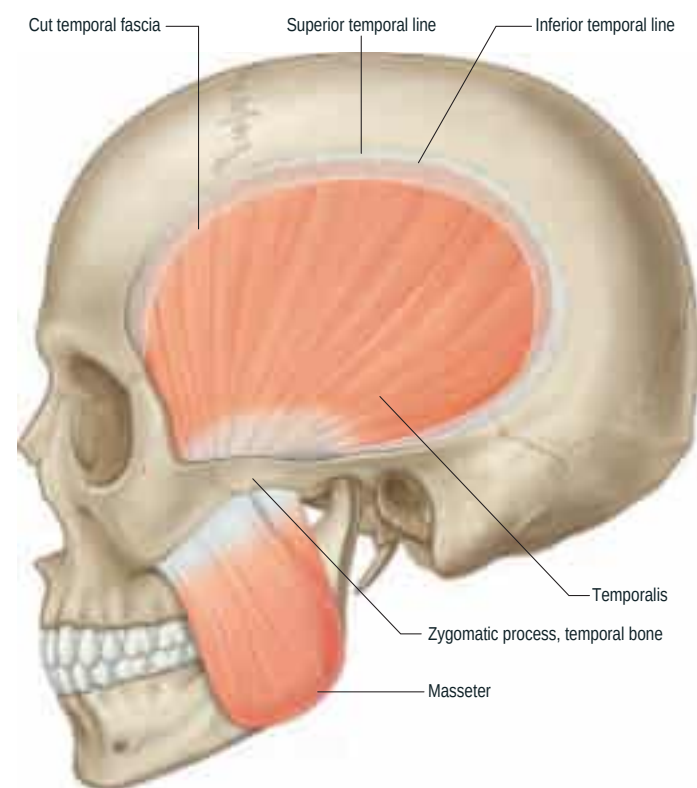


Fig. 32.18 Masseter and temporalis muscles. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Actions Masseter elevates the mandible to occlude the teeth in mastication and has a small effect in side-to-side movements, protraction and retraction. Its electrical activity in the resting position of the mandible is minimal.

Submasseteric space infections The submasseteric space is a potential space (see 'masticator space', p. 541) produced by the firm attachment of masseter to the ramus. Sometimes, infection around a mandibular third molar tooth tracks backwards, lateral to the mandibular ramus, and pus localizes deep to the attachment of masseter in the submasseteric tissue space. Such an abscess, lying deep to this thick muscle, produces little visible swelling but is accompanied by profound muscle spasm and severe limitation of jaw opening (trismus).

Temporalis

Temporalis (see Figs 32.18, 32.20) arises from the whole of the temporal fossa up to the inferior temporal line – except the part formed by the zygomatic bone – and from the deep surface of the temporal fascia. Its fibres converge and descend into a tendon that passes through the gap between the zygomatic arch and the side of the skull. A plane exists beneath the temporal fascia, which is attached to the superior surface of the zygomatic arch, and the muscle, which passes beneath the arch. An elevator introduced into this plane through an incision above the hairline may therefore be placed beneath a fractured zygomatic arch or bone in order to reduce the fracture (Gillies approach). Temporalis is attached to the medial surface, apex, anterior and posterior borders of the coronoid process and to the anterior border of the mandibular ramus almost up to the third molar tooth. Its anterior fibres are orientated vertically, the most posterior fibres almost horizontally, and the intervening fibres with intermediate degrees of obliquity, in the manner of a fan. Fibres of temporalis may occasionally gain attachment to the articular disc. Though less common than enlargement of masseter, hypertrophy of temporalis may develop unilaterally or bilaterally, producing prominence of the temporal fossa, which may be considered unsightly; botulinum toxin to paralyse the motor nerves is the preferred treatment for this condition.

Relations Skin, auriculares anterior and superior, temporal fascia, superficial temporal vessels, the auriculotemporal nerve, temporal branches of the facial nerve, the zygomaticotemporal nerve, the epicranial aponeurosis, the zygomatic arch and the masseter muscle are all superficial relations. Posterior relations of temporalis are the temporal

fossa above and the major components of the infratemporal fossa below. Behind the tendon of the muscle, the masseteric nerve and vessels traverse the mandibular notch. The anterior border is separated from the zygomatic bone by a mass of fat.

Vascular supply Temporalis is supplied by deep temporal branches from the second part of the maxillary artery, which enter on its deep aspect, and middle temporal branches from the superficial temporal artery, which enter on its lateral aspect. The anterior deep temporal artery supplies 20% of the muscle anteriorly; the posterior deep temporal artery, entering the mid-portion of the muscle, supplies 40% of the muscle in the mid-region; and the middle temporal artery, entering the muscle posteriorly, supplies 40% of the muscle in its posterior region. Considerable vascular anastomoses are present within the muscle (Cheung 1996).

Innervation Temporalis is supplied by the anterior, middle and posterior deep temporal branches of the anterior trunk of the mandibular nerve.

Actions Temporalis elevates the mandible and so closes the mouth and approximates the teeth. This movement requires both the upward pull of the anterior fibres and the backward pull of the posterior fibres because the head of the mandibular condyle rests on the articular eminence when the mouth is open. The muscle also contributes to side-to-side grinding movements. The posterior fibres retract the mandible after it has been protruded. The posterior fibres of temporalis, which are almost horizontal, are the only source of mandibular retrusion.

Lateral pterygoid

Lateral pterygoid (Fig. 32.20) is a short, thick muscle consisting of two parts. The upper head arises from the infratemporal surface and infratemporal crest of the greater wing of the sphenoid bone. The lower head arises from the lateral surface of the lateral pterygoid plate. From the two origins, the fibres converge, and pass backwards and laterally, to be inserted into a depression on the front of the neck of the mandible (the pterygoid fovea). A part of the upper head may be attached to the capsule of the temporomandibular joint and to the anterior and medial borders of its articular disc. Unlike the other muscles of mastication, lateral pterygoid is not pennate, nor does it have a significant number of Golgi tendon organs associated with its attachments.

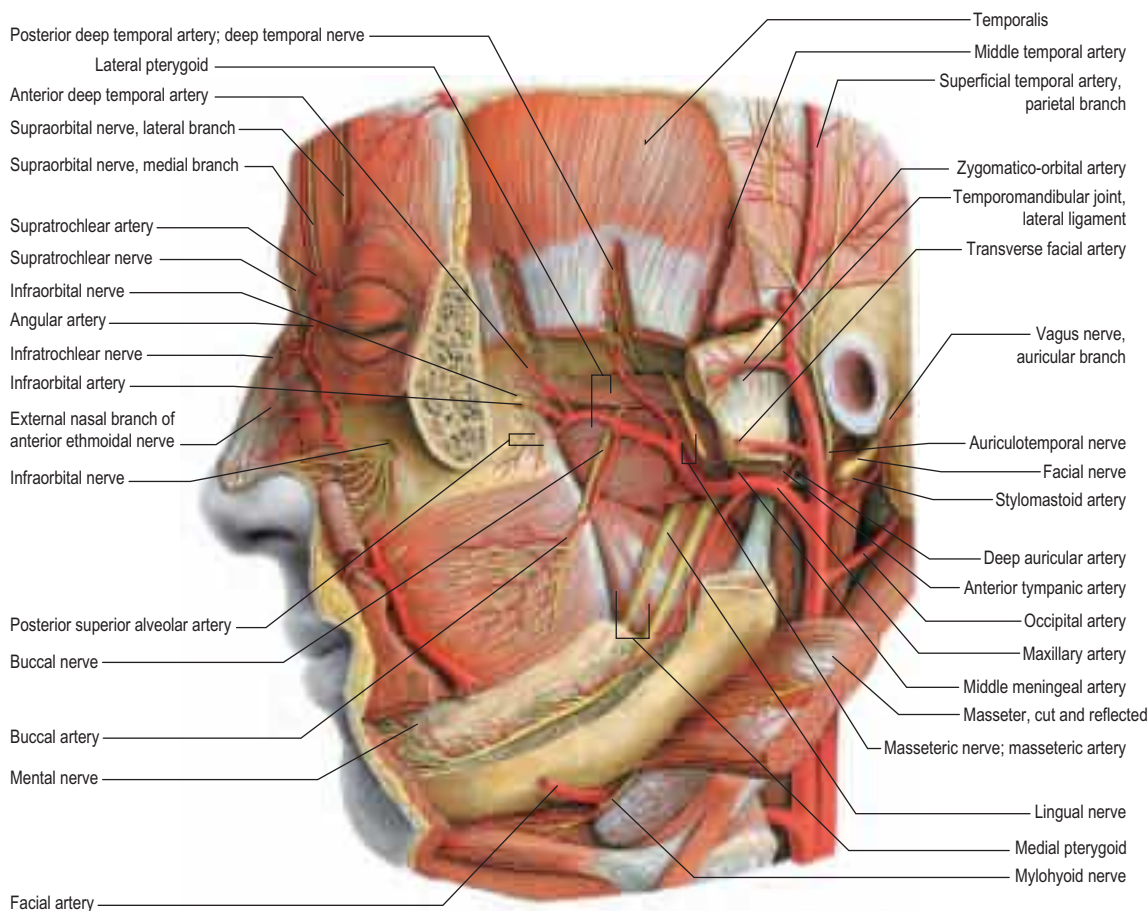


Fig. 32.20 Arteries and nerves of the head; deeper lateral regions. Most of the zygomatic arch and the ramus and body of the mandible have been removed; masseter has been cut and reflected laterally. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

The arterial supply is axial, i.e. the vessels run the entire length of the muscle. It is this vascular arrangement that permits the use of temporalis as an inferiorly based pedicled flap, either dynamic or static (increased length being obtained with division of the coronoid process), for the reconstruction of a variety of local facial and oral defects and in the treatment of facial paralysis. The temporalis fascia can be incorporated with the flap. The muscle may be split longitudinally (along its

long axis), retaining its axial supply from individual arteries; the vessels are identified at operation if necessary, with a Doppler. A further vascular advantage is that the arterial supply is disposed medially (deep temporal) and laterally (middle temporal), and there are few vessels in the mid-sagittal plane, which means that splitting of the muscle flap in the sagittal plane is possible (Fig. 32.19) (Cheung 1996).

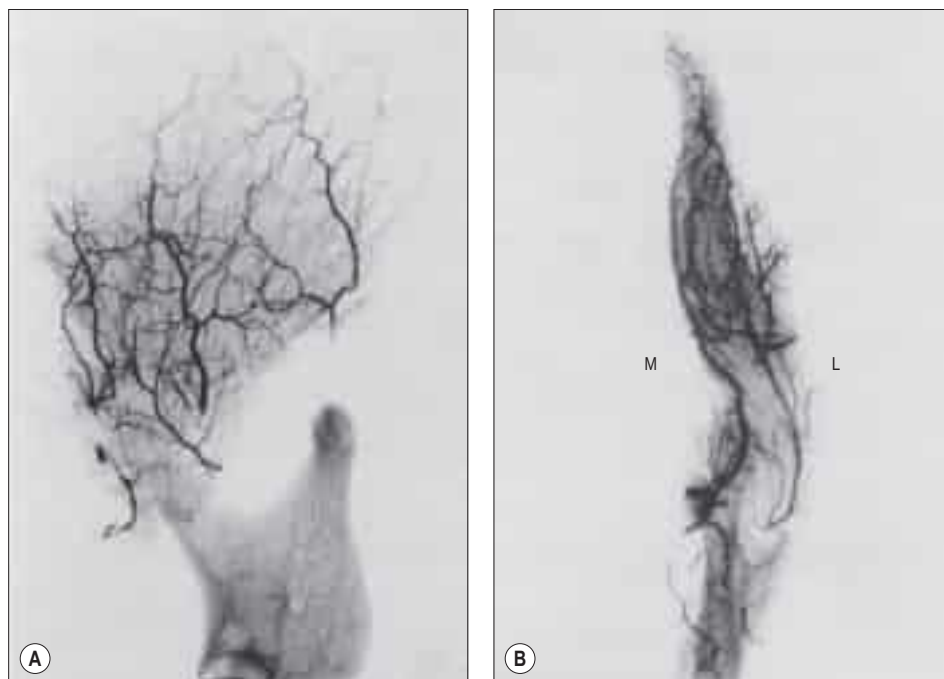


Fig. 32.19 The arterial network of the human temporalis. **A**, Lateral plane. **B**, Coronal plane. In the lateral plane, the muscle is supplied by the three primary arteries: the anterior deep temporal artery in the anterior portion, the posterior deep temporal artery in the mid-portion and the middle deep temporal artery in the posterior portion. In the coronal plane, the arterial distribution is concentrated mainly lateral (L) and medial (M) with less vascular density in the mid-sagittal region of the muscle. (With permission from Cheung LK. The vascular anatomy of the human temporalis muscle: implications for surgical splitting techniques. *Int J Oral Maxillofac Surg* 1996;25:414–421.)

Relations The mandibular ramus and masseter, the maxillary artery, which crosses either deep or superficial to the muscle, and the superficial head of medial pterygoid and the tendon of temporalis are all superficial relations. Deep to the muscle are the deep head of medial pterygoid, the sphenomandibular ligament, the middle meningeal artery and the mandibular nerve. The upper border is related to the temporal and masseteric branches of the mandibular nerve and the lower border is related to the lingual and inferior alveolar nerves. A deeply placed posterior superior alveolar nerve block has been known also to anaesthetize the lingual nerve. The buccal nerve and the maxillary artery pass between the two heads of the muscles. In temporomandibular joint dysfunction syndrome, spasm of lateral pterygoid can give rise to tenderness when palpating behind the maxillary tuberosity high in the buccal sulcus (the pterygoid sign).

Vascular supply Lateral pterygoid is supplied by pterygoid branches from the maxillary artery that are given off as the artery crosses the muscle and from the ascending palatine branch of the facial artery.

Innervation The nerves to lateral pterygoid (one for each head) arise from the anterior trunk of the mandibular nerve, deep to the muscle. The upper head and the lateral part of the lower head receive their innervation from a branch given off from the buccal nerve. However, the medial part of the lower head has a branch arising directly from the anterior trunk of the mandibular nerve.

Actions Lateral pterygoid has assumed a specialized role in mandibular opening that is mediated by its horizontally orientated fibres. When left and right muscles contract together, the condyle is pulled forwards and slightly downwards. This protrusive movement alone has little or no function except to assist in opening the jaw. Digastric and geniohyoid are the main jaw opening muscles; unlike lateral pterygoid, when acting alone, they rotate the jaw open, provided other muscles attached to the hyoid prevent it from being pulled forwards. If only one lateral pterygoid contracts, the jaw rotates about a vertical axis passing roughly through the opposite condyle and is pulled medially towards the opposite side. This contraction, together with that of the adjacent medial pterygoid (both attached to the lateral pterygoid plate), provides most of the strong medially directed component of the force used when grinding food between teeth of the same side. It is arguably the most important function of the inferior head of lateral pterygoid.

It is often stated that the upper head is used to pull the articular disc forwards when the jaw is opened. However, electromyographic studies (e.g. McNamara 1973, Juniper 1981) have demonstrated that the upper and lower heads are reciprocally innervated, so that the lower head contracts during mouth opening whilst the upper head relaxes, the situation reversing during closure. An explanation for this surprising activity is as follows (Osborn 1995). Most of the power of a clenching force is due to contractions of masseter and temporalis. The associated backward pull of temporalis is greater than the associated forward pull of (superficial) masseter, and so their combined jaw-closing action potentially pulls the condyle backwards. This is prevented by the simultaneous contraction of the upper head of lateral pterygoid, which stabilizes the condylar head against the articular eminence during closure, particularly during biting and mastication.

Medial pterygoid

Medial pterygoid (see Fig. 32.20) is a thick, quadrilateral muscle with two heads of origin. Mirroring masseter (which lies laterally), its fibres descend posteroinferiorly at an angle of about 10° to the vertical. In the coronal plane, it forms a 30° angle with the ramus. The major component is the deep head that arises from the medial surface of the lateral pterygoid plate of the sphenoid bone and is therefore deep to the lower head of lateral pterygoid. The small, superficial head arises from the maxillary tuberosity and the pyramidal process of the palatine bone, and therefore lies on the lower head of lateral pterygoid. The fibres of medial pterygoid descend posterolaterally and are attached by a strong tendinous lamina to the posteroinferior part of the medial surface of the ramus and angle of the mandible, as high as the mandibular foramen and almost as far forwards as the mylohyoid groove. This area of attachment is often ridged. Inferior alveolar nerve block injection can occasionally cause haemorrhage into the muscle, which may give rise to painful trismus. Medial pterygoid and masseter act together to support the angle of the mandible as the pterygomasseteric sling. Their strong muscle attachments resist lengthening of the mandibular ramus surgically.

Relations The lateral surface of medial pterygoid is related to the mandibular ramus, from which it is separated above its insertion by

lateral pterygoid, the sphenomandibular ligament, the maxillary artery, the inferior alveolar vessels and nerve, the lingual nerve and a process of the parotid gland. The medial surface is related to tensor veli palatini and is separated from the superior pharyngeal constrictor by styloglossus and stylopharyngeus and by some areolar tissue.

Vascular supply Medial pterygoid derives its main arterial supply from the pterygoid branches of the maxillary artery.

Innervation Medial pterygoid is innervated by the medial pterygoid branch of the mandibular nerve.

Actions The medial pterygoid muscles assist in elevating the mandible. Acting with the lateral pterygoids, they protrude it. When the medial and lateral pterygoids of one side act together, the corresponding side of the mandible is rotated both forwards and to the opposite side, with the opposite mandibular head as a vertical axis. Alternating activity in the left and right sets of muscles produces side-to-side movements, which are used to triturate food.

Pterygospinous ligament

The pterygospinous ligament, which is occasionally replaced by muscle fibres, stretches between the spine of the sphenoid bone and the posterior border of the lateral pterygoid plate near its upper end. It is sometimes ossified, and then completes a foramen that transmits the branches of the mandibular nerve to temporalis, masseter and lateral pterygoid.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Maxillary artery

The maxillary artery, the larger terminal branch of the external carotid artery, arises behind the neck of the mandible and is at first embedded in the parotid gland. It then crosses the infratemporal fossa to enter the pterygopalatine fossa through the pterygomaxillary fissure. The artery is widely distributed to the mandible, maxilla, teeth, muscles of mastication, palate, nose and cranial dura mater (see Fig. 32.20; Fig. 32.21). It will be described in three parts: mandibular, pterygoid and pterygopalatine.

The mandibular part runs horizontally by the medial surface of the ramus. It passes between the neck of the mandible and the sphenomandibular ligament, parallel with and slightly below the auriculotemporal nerve. It next crosses the inferior alveolar nerve and skirts the lower border of lateral pterygoid. The pterygoid part ascends obliquely forwards medial to temporalis and is usually superficial to the lower head of lateral pterygoid. When it runs deep to lateral pterygoid, it lies between the muscle and branches of the mandibular nerve, and may project as a lateral loop between the two parts of lateral pterygoid. Asymmetry in this pattern of distribution may occur between the right and left infratemporal fossae, and ethnic differences have been reported. Where the maxillary artery runs superficial to the lower head of lateral pterygoid, the most common pattern is that the artery passes lateral to the inferior alveolar, lingual and buccal nerves. Less frequently, only the buccal nerve crosses the artery laterally, and rarely the artery passes deep to all the branches of the mandibular nerve. The pterygopalatine part passes between the two heads of lateral pterygoid to reach the pterygomaxillary fissure before it passes into the pterygopalatine fossa, where it terminates as the third part of the maxillary artery.

The mandibular part of the maxillary artery has five branches that all enter bone, namely: deep auricular, anterior tympanic, middle meningeal, accessory meningeal and inferior alveolar arteries. The pterygoid part of the maxillary artery has five branches that do not enter bone but supply muscle, and include deep temporal, pterygoid, masseteric and buccal arteries. The branches of the pterygopalatine part of the artery accompany similarly named branches of the maxillary nerve (including those associated with the pterygopalatine ganglion) and are described on page 552.

Deep auricular artery

The deep auricular artery pierces the osseous or cartilaginous wall of the external acoustic meatus and supplies the skin of the external acoustic meatus and part of the tympanic membrane. A small branch contributes to the arterial supply of the temporomandibular joint.

Anterior tympanic artery

The anterior tympanic artery passes through the petrotympanic fissure to supply part of the lining of the middle ear and accompanies the chorda tympani nerve.

A

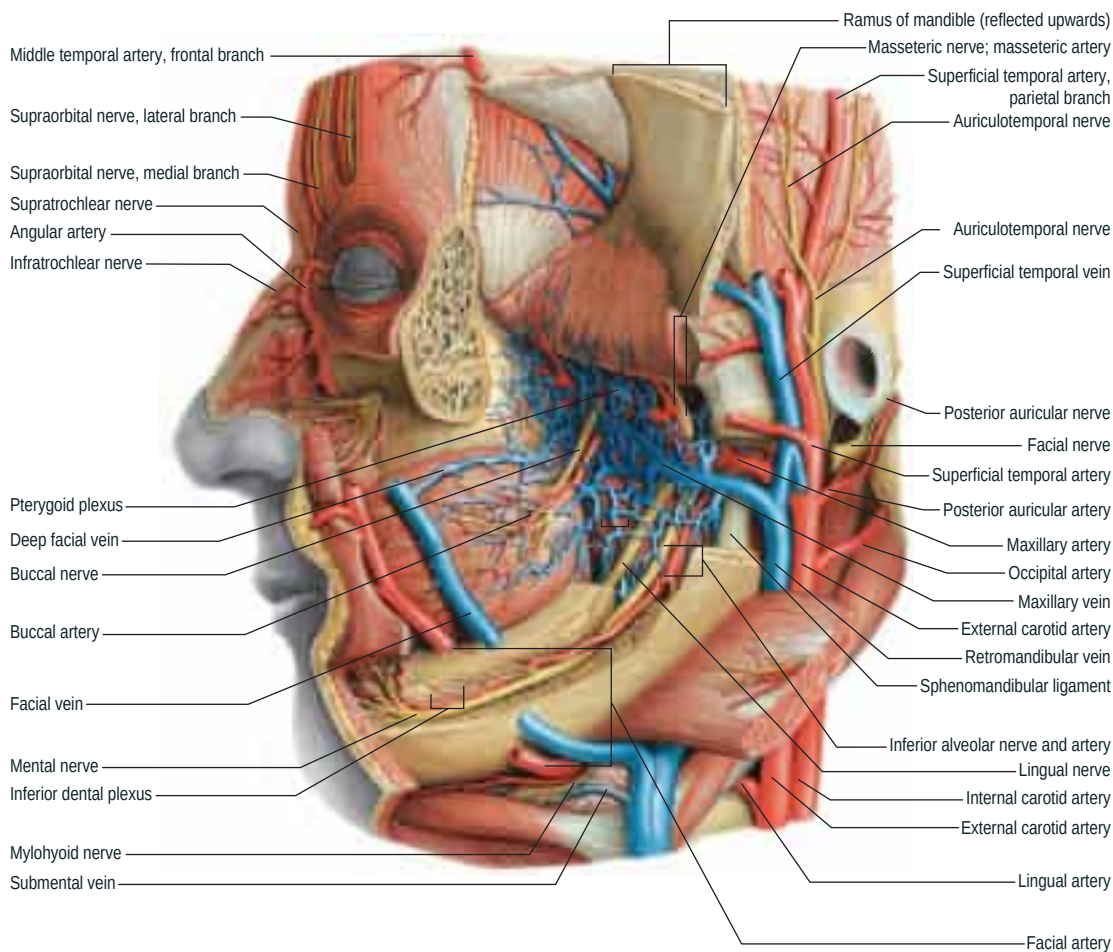
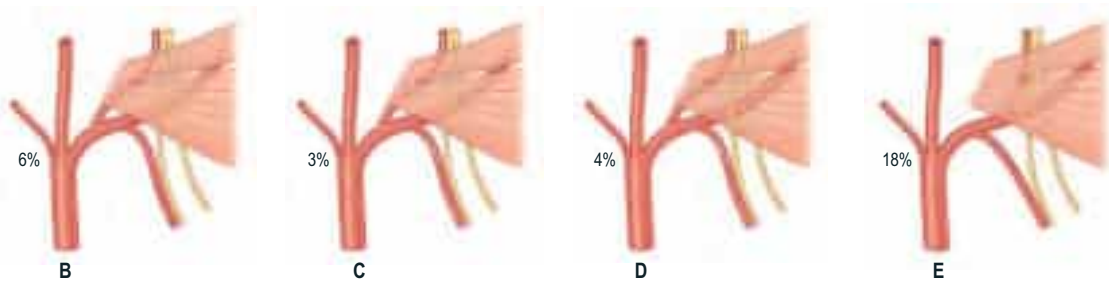


Fig. 32.21 A, Vessels and nerves of the head, exposed at a level deeper than that shown in [Figure 32.20](#). Note the pterygoid venous plexus. **B–E**, Variations in the course of the maxillary artery. **B**, The maxillary artery passes medial to lateral pterygoid, and to the lingual and inferior alveolar nerves. **C**, The artery passes between the lingual and inferior alveolar nerves. **D**, The artery passes through a loop formed by the inferior alveolar nerve. **E**, The middle meningeal artery branches off distal to the inferior alveolar artery. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)



Middle meningeal artery

The middle meningeal artery is the main source of blood to the bones of the vault of the skull ([Fig. 32.22](#)). It may arise either directly from the first part of the maxillary artery or from a common trunk with the inferior alveolar artery. When the maxillary artery lies superficial to lateral pterygoid, the middle meningeal artery is usually the first branch of the maxillary artery. However, when the maxillary artery takes a deep course in relation to the muscle, this is not usually the case. The middle meningeal artery ascends between the sphenomandibular ligament and lateral pterygoid, passes between the two roots of the auriculotemporal nerve, and leaves the infratemporal fossa through the foramen spinosum to enter the cranial cavity medial to the midpoint of the zygomatic bone. Its further course is described on [page 439](#).

Accessory meningeal artery

The accessory meningeal artery runs through the foramen ovale into the middle cranial fossa and may arise directly from the maxillary artery or as a branch of the middle meningeal artery itself. In its course in the infratemporal fossa, the accessory meningeal artery is closely related to tensor and levator veli palatini and usually runs deep to the mandibular nerve. Although it runs intracranially, its main distribution is extracranial, principally to medial pterygoid, lateral pterygoid (upper head), tensor veli palatini, the greater wing and pterygoid processes of the sphenoid, branches of the mandibular nerve and the otic ganglion. The accessory meningeal artery is sometimes replaced by separate small arteries.

Inferior alveolar artery

The inferior alveolar artery descends in the infratemporal fossa posterior to the inferior alveolar nerve, between the ramus laterally and the sphenopalatine ligament medially. Two anterior branches are given off the artery prior to its entry into the mandibular foramen. The further course of the artery is described on [page 526](#).

Deep temporal arteries

The arterial supply to the temporalis in the coronal plane is concentrated mainly on its medial and lateral aspects. The anterior and posterior branches of the deep temporal arteries pass between temporalis and the pericranium, producing shallow grooves in the bone. They anastomose with the middle temporal branch of the superficial temporal artery situated laterally ([Cheung 1996](#)). The anterior deep temporal artery connects with the lacrimal artery by small branches which perforate the zygomatic bone and greater wing of the sphenoid.

Masseteric artery

The masseteric artery, which is small, accompanies the masseteric nerve as it passes behind the tendon of temporalis through the mandibular incisure (notch) to enter the deep surface of masseter (see [Fig. 32.20](#)). Its branches can also supply the temporomandibular joint. The masseteric artery anastomoses with the masseteric branches of the facial artery and with the transverse facial branch of the superficial temporal artery.

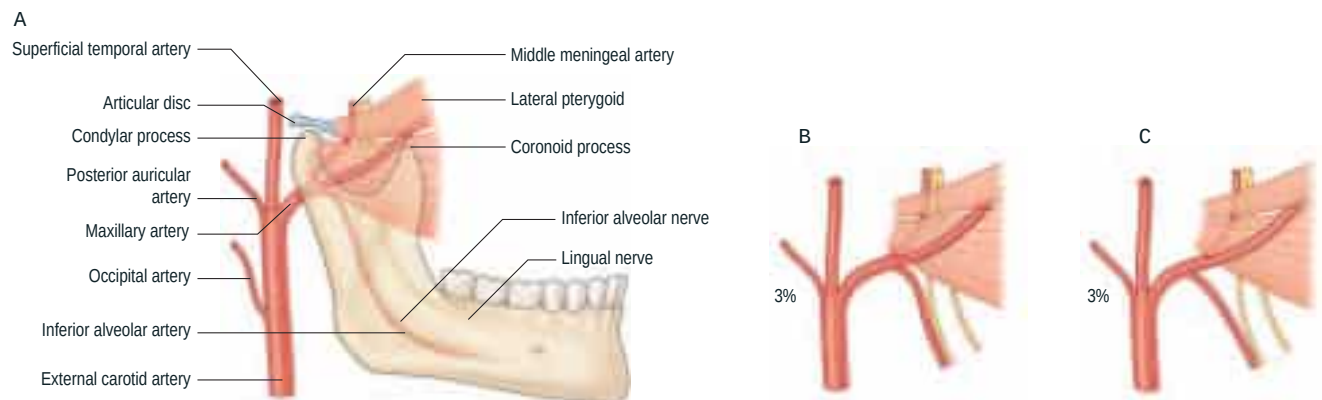


Fig. 32.22 Variations in the origin of the middle meningeal artery. **A**, The middle meningeal artery branches off proximal to the inferior alveolar artery. **B**, The artery branches off opposite the inferior alveolar artery. **C**, The artery branches off distal to the inferior alveolar artery. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Pterygoid arteries

The pterygoid arteries are irregular in number and origin, and are distributed to lateral and medial pterygoid.

Buccal artery

The buccal artery runs obliquely forwards between medial pterygoid and the attachment of temporalis, and supplies the skin and mucosa over buccinator, accompanying the lower part of the buccal branch of the mandibular nerve (see Fig. 32.21A). It anastomoses with branches of the facial and infraorbital arteries. A small lingual branch may be given off to accompany the lingual nerve and supply structures in the floor of the mouth.

Maxillary veins and the pterygoid venous plexus

Maxillary vein

The maxillary vein is a short trunk that accompanies the first part of the maxillary artery (see Fig. 32.21A). It is formed from the confluence of veins from the pterygoid plexus and passes back between the sphenomandibular ligament and the neck of the mandible, to enter the parotid gland. It unites within the substance of the gland with the superficial temporal vein to form the retromandibular vein.

Pterygoid venous plexus

The pterygoid plexus of veins consists of copious large-calibre veins surrounding the pterygoid segment of the maxillary artery. It is found partly between temporalis and lateral pterygoid, and partly between the two pterygoid muscles. Sphenopalatine, deep temporal, pterygoid, masseteric, buccal, alveolar (dental), greater palatine and middle meningeal veins and a branch or branches from the inferior ophthalmic vein are all tributaries. The plexus connects with the facial vein via the deep facial vein, with the cavernous sinus through veins that pass through the sphenoidal emissary foramen (of Vesalius), foramen ovale and foramen lacerum and with the orbit via the inferior ophthalmic vein(s). Its deep temporal tributaries often connect with tributaries of the anterior diploic veins and thus with the middle meningeal veins.

Lymphatic drainage

Lymphatic drainage of the infratemporal fossa (and the temporal fossa) is to the superior deep cervical nodes.

INNERVATION

The infratemporal fossa contains the major subdivisions of the mandibular branch of the trigeminal nerve, together with the chorda tympani, which enters the fossa and joins the lingual nerve, and the otic ganglion, which is functionally related to the parotid gland. The main sensory branches of the mandibular nerve extend beyond the infratemporal fossa and their distribution to the face is described on page 502.

Mandibular nerve

The mandibular nerve is the largest trigeminal division and is a mixed nerve. Its sensory branches supply the teeth and gums of the mandible,

the skin in the temporal region, part of the auricle – including the external meatus and tympanic membrane – and the lower lip, the lower part of the face and the mucosa of the anterior two-thirds (presulcal part) of the tongue and the floor of the oral cavity (see Figs 32.20, 32.21A; Fig. 32.23). The motor branches innervate the muscles of mastication. The large sensory root emerges from the lateral part of the trigeminal ganglion and exits the cranial cavity through the foramen ovale. The small motor root passes under the ganglion and through the foramen ovale to unite with the sensory root just outside the skull. As it descends from the foramen ovale, the nerve is usually around 4 cm from the surface and a little anterior to the neck of the mandible. The mandibular nerve immediately passes between tensor veli palatini, which is medial, and lateral pterygoid, which is lateral, and gives off a meningeal branch and the nerve to medial pterygoid from its medial side. The nerve then divides into a small anterior and large posterior trunk. The anterior division gives off branches to the four main muscles of mastication and a buccal branch that is sensory to the cheek. The posterior division gives off three main sensory branches – the auriculotemporal, lingual and inferior alveolar nerves – and motor fibres that supply mylohyoid and the anterior belly of digastric.

As is the case with the extraocular muscles, the masticatory muscles function as a group; only when pathology intervenes do their individual actions, or lack of action, become apparent. Wasting of the masticatory muscles, most evident in the temporal fossa, happens when tumours affect the motor root of the mandibular nerve. The mandible deviates to the side of the pathology as a result of the unopposed action of the contralateral lateral and medial pterygoids.

Meningeal branch (nervus spinosus)

The meningeal branch re-enters the cranium through the foramen spinosum with the middle meningeal artery. It divides into anterior and posterior branches, which accompany the main divisions of the middle meningeal artery and supply the dura mater in the middle cranial fossa and, to a lesser extent, in the anterior fossa and calvarium.

Nerve to medial pterygoid

The nerve to medial pterygoid is a slender ramus that enters the deep aspect of the muscle. It supplies one or two filaments that pass through the otic ganglion without interruption to supply tensor tympani and tensor veli palatini.

Anterior trunk of mandibular nerve

The anterior trunk of the mandibular nerve gives rise to the buccal nerve, which is sensory, and the masseteric, deep temporal and lateral pterygoid nerves, which are all motor.

Buccal nerve

The buccal nerve passes between the two heads of lateral pterygoid (see Fig. 32.23). It descends deep to the tendon of temporalis, passes laterally in front of masseter, and anastomoses with the buccal branches of the facial nerve. It carries the motor fibres to lateral pterygoid, and these are given off as the buccal nerve passes through the muscle. It may also give off the anterior deep temporal nerve. The buccal nerve supplies sensation to the skin over the anterior part of buccinator and the buccal mucous membrane, together with the posterior part of the buccal gingivae adjacent to the second and third molar teeth.

Nerve to masseter

The nerve to masseter passes laterally above lateral pterygoid, on to the skull base, anterior to the temporomandibular joint and posterior to

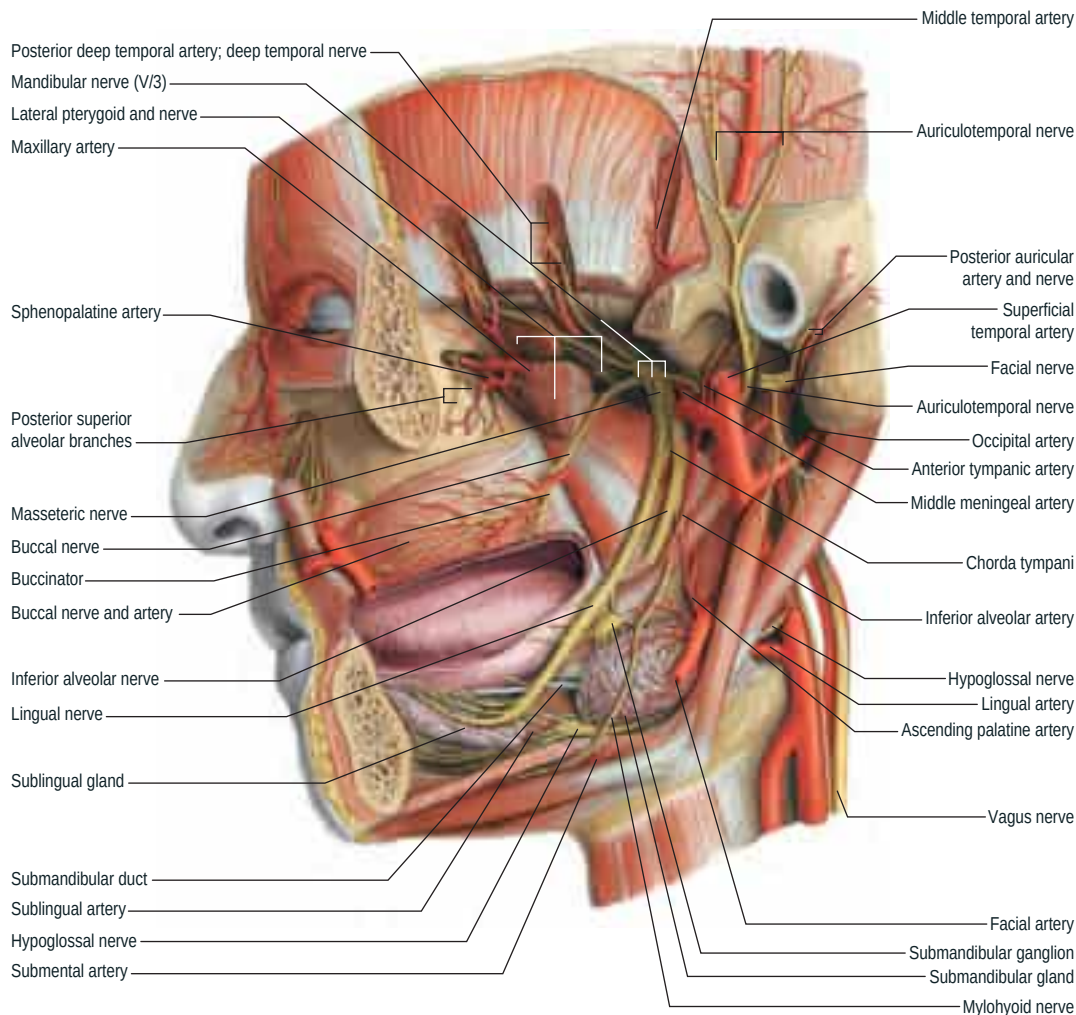


Fig. 32.23 Arteries and nerves of the head, deepest lateral regions. Compare with [Figures 32.20](#) and [32.21A](#). (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

the tendon of temporalis. It crosses the posterior part of the mandibular notch with the masseteric artery and ramifies on and enters the deep surface of masseter. It also provides articular branches that supply the temporomandibular joint.

Deep temporal nerves

The deep temporal nerves usually consist of two branches, anterior and posterior, although there may be a middle branch. They pass above lateral pterygoid to enter the deep surface of temporalis. The anterior nerve frequently arises as a branch of the buccal nerve. The small posterior nerve sometimes arises in common with the nerve to masseter.

Nerve to lateral pterygoid

The nerve to lateral pterygoid enters the deep surface of the muscle. It may arise separately from the anterior division of the mandibular nerve or from the buccal nerve.

Posterior trunk of mandibular nerve

The posterior trunk of the mandibular nerve is larger than the anterior and is mainly sensory, although it receives fibres from the motor root for the nerve to mylohyoid. It divides into auriculotemporal, lingual and inferior alveolar (dental) nerves (see [Fig. 32.20A](#)).

Auriculotemporal nerve

The auriculotemporal nerve usually has two roots that encircle the middle meningeal artery. It runs back under lateral pterygoid on the surface of tensor veli palatini, passes between the sphenomandibular ligament and the neck of the mandible, and then runs laterally behind the temporomandibular joint related to the upper part of the parotid gland. Emerging from behind the joint, it ascends over the posterior root of the zygoma, posterior to the superficial temporal vessels, and divides into superficial temporal branches. It communicates with the facial nerve and otic ganglion. The rami to the facial nerve, usually two, pass anterolaterally behind the neck of the mandible to join the facial nerve at the posterior border of masseter. Filaments from the otic ganglion join the roots of the auriculotemporal nerve close to their origin. The sensory distribution of the auriculotemporal nerve on the face is described on [page 502](#).

Lingual nerve

The lingual nerve is sensory to the mucosa of the anterior two-thirds of the tongue, the floor of the mouth and the mandibular lingual gingivae. It arises from the posterior trunk of the mandibular nerve and at first runs beneath lateral pterygoid and superficial to tensor veli palatini, where it is joined by the chorda tympani branch of the facial nerve, and often by a branch of the inferior alveolar nerve. Emerging from under cover of lateral pterygoid, the lingual nerve then runs downwards and forwards on the surface of medial pterygoid (see [Fig. 32.23](#)), and is thus carried progressively closer to the medial surface of the mandibular ramus. It becomes intimately related to the bone a few millimetres below and behind the junction of the vertical ramus and horizontal body of the mandible. Here it lies anterior to, and slightly deeper than, the inferior alveolar (dental) nerve. It next passes below the mandibular attachment of the superior pharyngeal constrictor and pterygomandibular raphe, closely applied to the periosteum of the medial surface of the mandible, until it lies opposite the posterior root of the third molar tooth, where it is covered only by the gingival mucoperiosteum. At this point, it usually lies approximately 3 mm below the alveolar crest with a mean distance from the bone of 2 mm. Significantly, in more than 25% of cases, the nerve may be in direct contact with the bone at this site, but sometimes it lies above the alveolar crest. It next passes medial to the mandibular origin of mylohyoid, and this carries it progressively away from the mandible, and separates it from the alveolar bone covering the mesial root of the third molar tooth ([Pogrel et al 1995](#)). The rest of the nerve is described with the mouth and oral cavity on [page 514](#).

Inferior alveolar (dental) nerve

The inferior alveolar nerve descends behind lateral pterygoid. At the lower border of the muscle, the nerve passes between the sphenomandibular ligament and the mandibular ramus, and enters the mandibular canal via the mandibular foramen ([Barker and Davies 1972](#)). Below lateral pterygoid, it is accompanied by the inferior alveolar artery (see [Fig. 32.23](#)), a branch of the first part of the maxillary artery, which also enters the canal with associated veins. The subsequent course of the inferior alveolar nerve is described on [page 527](#).

The anterior deep temporal nerve may arise from the buccal nerve, and the posterior deep temporal nerve may arise from the nerve to masseter. In addition, branches from the buccal and masseteric nerves may supply the inferior aspect of the muscle independently from the deep temporal nerves. Discrete neuromuscular subunits have been identified within the muscle based on the anterior, middle and posterior deep temporal nerves, suggesting the possibility of individual dynamic muscle flaps, each with a different vector, in the treatment of facial palsy (Chang et al 2013) (Fig. 32.24).

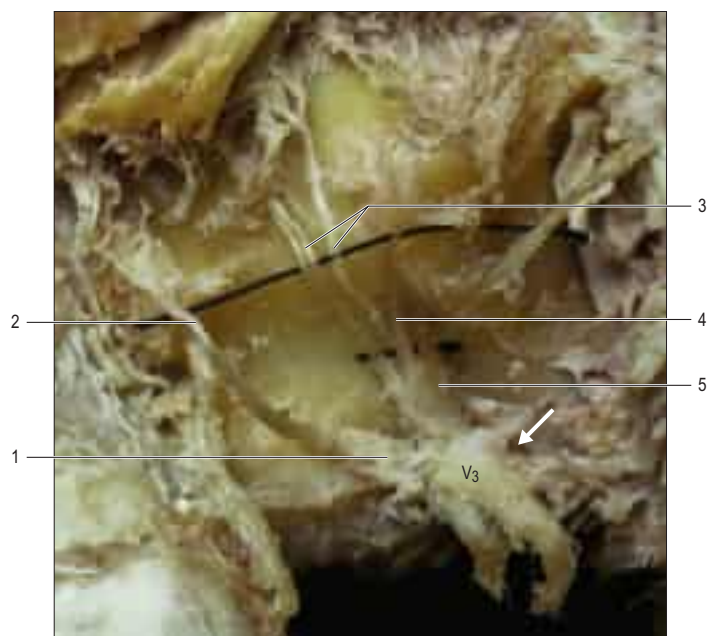


Fig. 32.24 The foramen ovale, inferolateral view. Nerves supplying temporalis originate from the anterior trunk of the mandibular division of the trigeminal nerve V3 (arrow). Key: 1, (long) buccal nerve; 2, anterior deep temporal nerve; 3, middle deep temporal nerve; 4, posterior deep temporal nerve; 5, nerve to masseter. (Courtesy of David Cantelmi, University of Toronto, Canada. With permission from Chang Y, Cantelmi D, Wisco J, et al. Evidence for the functional compartmentalization of the temporalis muscle: a 3-dimensional study of innervation. *J Oral Maxillofac Surg* 2013;71:1170–1177.)

Neural anastomoses with the mandibular nerve



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Otic ganglion

The otic ganglion is a small (major diameter 3–4 mm), oval, flat reddish-grey ganglion situated just below the foramen ovale. It is a peripheral parasympathetic ganglion related topographically to the mandibular nerve but connected functionally with the glossopharyngeal nerve (see Fig. 37.13). Near its junction with the trigeminal motor root, the mandibular nerve lies lateral to the ganglion; tensor veli palatini lies medially, separating the ganglion from the cartilaginous part of the pharyngotympanic tube, and the middle meningeal artery is posterior to the ganglion. The otic ganglion usually surrounds the origin of the nerve to medial pterygoid.

Like all parasympathetic ganglia, there are three roots: motor, sympathetic and sensory. Only the parasympathetic fibres relay in the ganglion (see Fig. 26.7). The motor, parasympathetic, root of the otic ganglion is the lesser petrosal nerve, conveying preganglionic fibres from the glossopharyngeal nerve, which originate from neurones in the inferior salivatory nucleus. The lesser petrosal nerve runs intracranially in the middle cranial fossa on the anterior surface of the petrous bone before passing through either the foramen ovale or the sphenopetrosal fissure or the innominate canal of Arnold (a canal in the greater wing of the sphenoid close to the foramen spinosum), to join the otic ganglion. The nerve synapses in the otic ganglion, and postganglionic fibres pass by a communicating branch to the auriculotemporal nerve and so to the parotid gland. The sympathetic root arises from a plexus on the middle meningeal artery. It contains postganglionic fibres from the superior cervical sympathetic ganglion, which traverse the otic ganglion without relay and emerge with parasympathetic fibres in the connection with the auriculotemporal nerve to supply blood vessels in the parotid gland. The sensory fibres from the gland are derived from the auriculotemporal nerve. Clinical observations suggest that in humans the gland also receives secretomotor fibres through the chorda tympani.

Branches

A branch connects the otic ganglion to the chorda tympani nerve, while another ramus ascends to join the nerve of the pterygoid canal. These branches may form an additional pathway by which gustatory fibres from the anterior two-thirds of the tongue may reach the facial ganglion without traversing the middle ear, and they do not synapse in the otic ganglion. Motor branches to tensor veli palatini and tensor tympani, derived from the nerve to medial pterygoid, also pass through the ganglion without synapsing. (For further reading, see Senger et al (2014).)

Chorda tympani

The chorda tympani nerve enters the infratemporal fossa region by passing through the medial end of the petrotympanic fissure behind the capsule of the temporomandibular joint. The nerve descends medial to the spine of the sphenoid bone – which it sometimes grooves – lying posterolateral to tensor veli palatini. It is crossed medially by the middle meningeal artery, the roots of the auriculotemporal nerve and by the inferior alveolar nerve (see Fig. 32.23). The chorda tympani joins the posterior aspect of the lingual nerve at an acute angle. It carries taste fibres for the anterior two-thirds of the tongue and efferent preganglionic parasympathetic (secretomotor) fibres destined for the submandibular ganglion in the floor of the mouth.

SPREAD OF INFECTION FROM THE INFRATEMPORAL FOSSA

Infections from most teeth in the upper jaw or the front of the lower jaw usually drain harmlessly into the oral cavity, either via the vestibule buccally, or via the palate or mouth lingually. In contrast, a pericoronitis that affects a mandibular third molar tooth that communicates with the oral cavity, or, less commonly, either a dental abscess associated with this tooth, or a postoperative infection, may spread along tissue planes. Spread may be buccal; submandibular; submental; parapharyngeal; sublingual; less commonly, submasseteric; and, more rarely, infratemporal. The main symptom caused by infection of the pterygomandibular region is trismus (painful reflex muscle spasm), which usually affects medial pterygoid. There are few anatomical barriers to

the spread of infection between neighbouring tissue spaces from the pterygomandibular space. When the submandibular, sublingual and submental areas are affected either unilaterally or bilaterally (bilateral involvement is known as Ludwig's angina), there is a risk that the airway may become obstructed. Infection may (potentially) spread further, usually to the parapharyngeal, and occasionally to the retropharyngeal, spaces or the tissue planes of the face above, when it may even reach the orbit via the inferior orbital fissure. Life-threatening cavernous sinus thrombosis is a remote possibility once infection has spread to the orbit, with further spread directly through the superior orbital fissure into the cranial cavity. Infection may also spread to the cavernous sinus from the pterygoid plexus and ophthalmic veins (Smyth 2007, Guidera et al 2014) (Fig. 32.26).

PTERYGOPALATINE FOSSA

The pterygopalatine fossa is a small pyramidal space below the apex of the orbit on the lateral side of the skull. Placed between the infratemporal fossa laterally and the nasopharynx medially, it functions as a neurovascular conduit. The body of the sphenoid forms the roof, the posterior boundary is the root of the pterygoid process and adjoining anterior surface of the greater wing of the sphenoid, and the anterior boundary is the superomedial part of the infratemporal surface (posterior wall) of the maxilla. The perpendicular plate of the palatine bone, with its orbital and sphenoidal processes, forms the medial boundary, and the pterygomaxillary fissure is the lateral boundary. There are two openings in the posterior wall of the pterygopalatine fossa: the foramen rotundum, which transmits the maxillary nerve, and the pterygoid canal (Vidian canal), which transmits the nerve of the pterygoid canal (Vidian nerve). When the anterior aspect of the pterygoid plate is examined in a disarticulated sphenoid, the foramen rotundum is seen to lie above and lateral to the pterygoid canal (see Figs 32.3A, B).

The fossa communicates with the nasal cavity via the sphenopalatine foramen; with the orbit via the medial end of the inferior orbital fissure; and with the infratemporal fossa via the pterygomaxillary fissure, which lies between the back of the maxilla and the pterygoid process of the sphenoid and transmits the maxillary artery. It also communicates with the middle fossa via foramen rotundum and the pterygoid canal, and with the oral cavity via the greater palatine canal, which opens in the posterolateral aspect of the hard palate. Its proximity to these adjacent areas permits the ready spread of both tumours and infection.

The main contents of the pterygopalatine fossa are the third part of the maxillary artery, the maxillary nerve and many of its branches, and the pterygopalatine ganglion.

Maxillary artery

The maxillary artery passes through the pterygomaxillary fissure from the infratemporal fossa into the pterygopalatine fossa, where it terminates as the third part of the maxillary artery. The artery has a variable and tortuous course in its short passage through the pterygopalatine fossa, where it gives off numerous branches, including the posterior superior alveolar and infraorbital arteries and the artery of the pterygoid canal (Vidian artery), and terminates in the sphenopalatine and greater palatine arteries.

Posterior superior alveolar artery

The posterior superior alveolar artery arises from the maxillary artery within the pterygopalatine fossa and runs through the pterygomaxillary fissure on to the maxillary tuberosity. It gives off branches that penetrate the bone here to supply the maxillary molar and premolar teeth and the maxillary air sinus, and other branches that supply the buccal mucosa. Occasionally, the posterior superior alveolar artery arises from the infraorbital artery.

Infraorbital artery

The infraorbital artery enters the orbit through the inferior orbital fissure. It runs on the floor of the orbit in the infraorbital groove and infraorbital canal, and emerges on to the face at the infraorbital foramen to supply the lower eyelid, part of the cheek, the side of the external nose, and the upper lip. While within the infraorbital canal, it gives off the anterior superior alveolar artery, which runs downwards to supply the anterior teeth and the anterior part of the maxillary sinus. A middle superior alveolar artery is also often described. When present, it branches from the infraorbital artery within the infraorbital canal and runs inferiorly along the lateral wall of the maxillary sinus towards the region of the canine and lateral incisor teeth, anastomosing with the anterior and posterior superior alveolar arteries.

Extensive anastomoses, which may be sensory-to-sensory, sensory-to-motor or motor-to-motor, exist between the lower cranial nerves and upper cervical nerves. Variable anastomoses exist between all three of the divisions of the trigeminal nerve with branches of the facial nerve. The connection between the lingual nerve and the chorda tympani branch of the facial is the most familiar interconnection with the mandibular nerve. Connections have almost always been demonstrated between the auriculotemporal nerve and branches of the facial nerve (Fig. 32.25). The buccal branch may communicate with the buccal and zygomatic branches of the facial via the 'communicating buccal nerve'. Interconnections between the mental nerve and the buccal and marginal mandibular branch of the facial have been demonstrated. It is suggested that one possible explanation for these trigeminal-facial interconnections relates to the proprioceptive innervation of the facial musculature. A branch from the auriculotemporal may join the inferior alveolar nerve within the infratemporal fossa with evident implications for referred pain. The functional significance of some of the neural connections has yet to be determined (Shoja et al 2014).

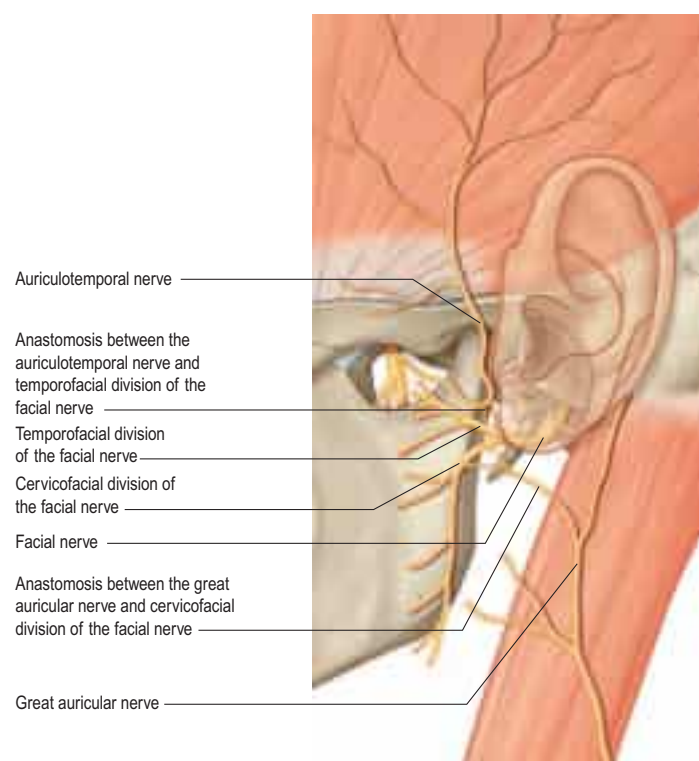


Fig. 32.25 Anastomoses of the auriculotemporal nerve with the temporofacial division of the facial nerve, and of the great auricular nerve with the cervicofacial division of the facial nerve in the parotid and periparotid regions. (With permission from Shoja MM, Oyesiku NM, Grissner CJ et al. Anastomoses between lower cranial and upper cervical nerves: a comprehensive review with potential significance during skull base and neck operations, Part 1: Trigeminal, facial and vestibulocochlear nerves. Clin Anat. 2014;27(1):118–30.)

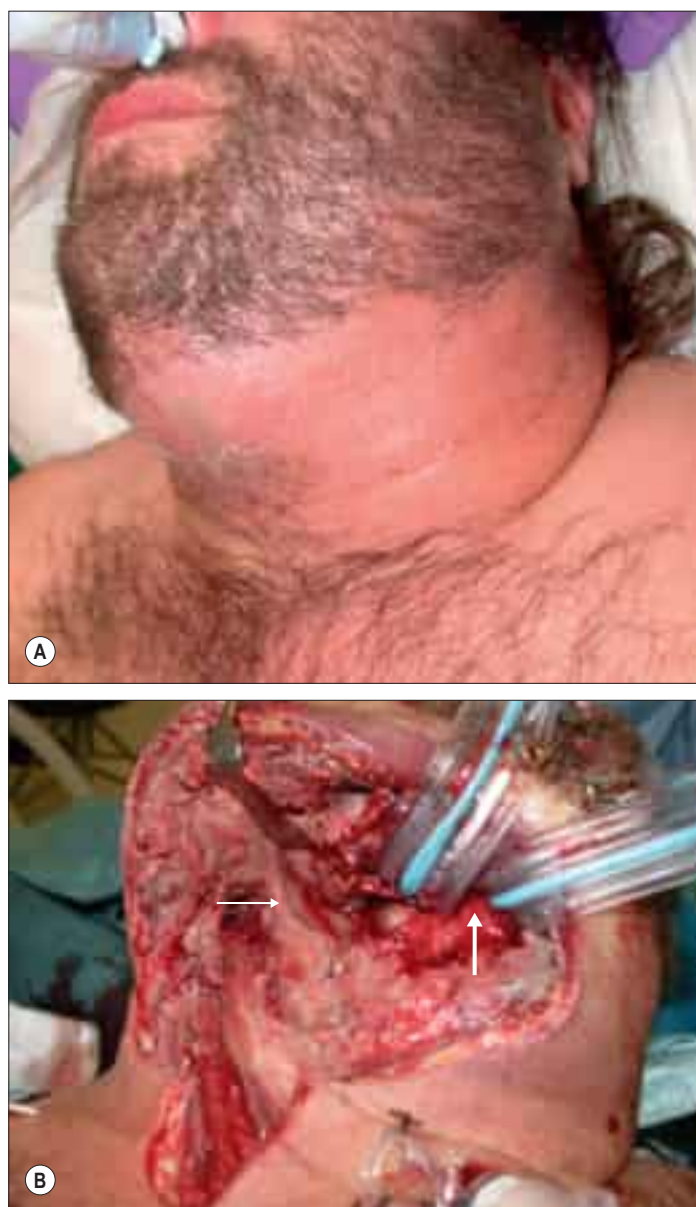


Fig. 32.26 **A**, Ludwig's angina: spread of infection from an infected lower left third molar to the pterygomandibular space and thence into the submandibular, sublingual and submental spaces bilaterally. The infection is life-threatening due to restricted mouth opening and a compromised airway. **B**, In this case, the airway was secured with awake fiberoptic intubation followed by tracheostomy. Aggressive through-and-through drainage was initiated (note the grey discoloration of the tissues). Prior to the availability of antibiotics, Ludwig's angina had a mortality rate of 75%. Aggressive early treatment with surgical drainage and intravenous antibiotics has reduced the mortality to approximately 5%. The narrow white arrow indicates the cut edge of mylohyoid. The broad white arrow indicates through-and-through corrugated drains. (Smyth AG 2007 Management of orofacial infections. In: Ward-Booth P, Schendel SA, Hausamen JE (eds) Maxillofacial Surgery, 2nd ed. Ch. 80, pp. 1550–71.)

Some authors describe anterior and posterior ‘compartments’ of the pterygopalatine fossa, where the anterior compartment contains the third part of the maxillary artery and its branches, and the posterior compartment contains the maxillary division of the trigeminal nerve and the nerve of the pterygoid canal, together with the pterygopalatine ganglion and its branches (Figs 32.27–32.28) (Alfieri et al 2003, Cavallo et al 2005, Falcon et al 2010).

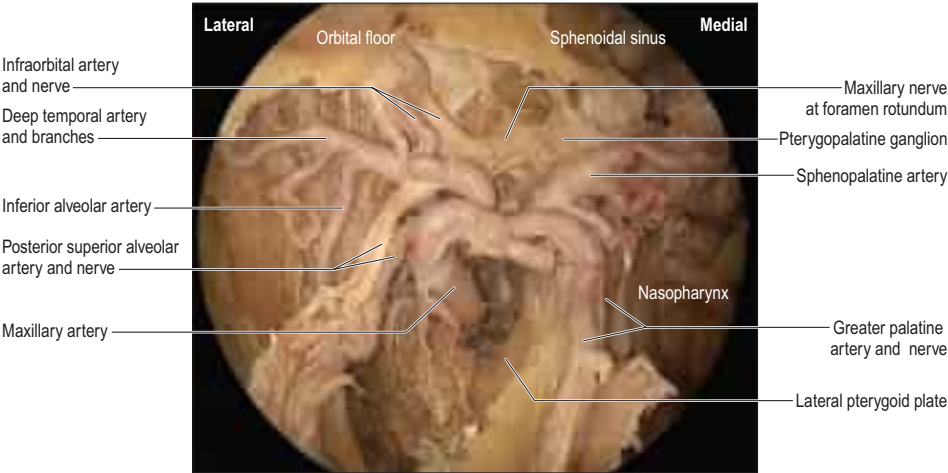


Fig. 32.27 An endoscopic view demonstrating the anterior (vascular) compartment of the right pterygopalatine fossa (00 lens endoscope). The vascular structures are anterior to the neural structures. Fat has been removed. (Image courtesy of Professor C Snyderman, University of Pittsburgh Medical Center.)

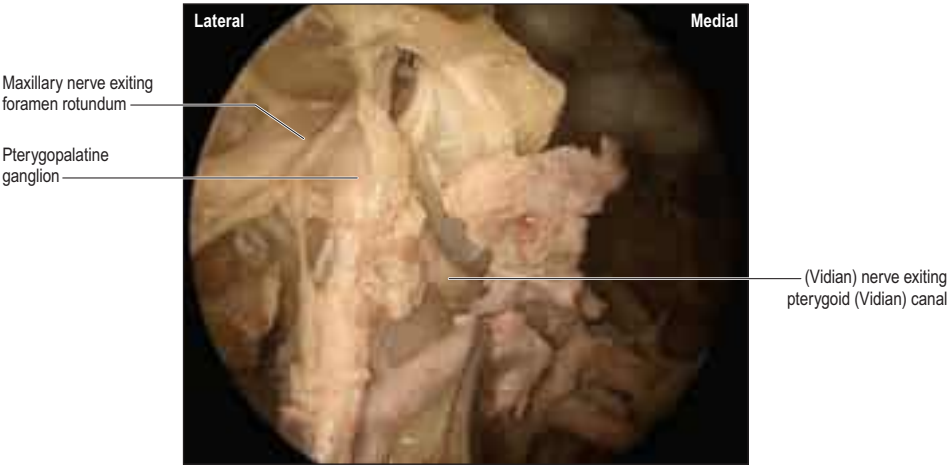


Fig. 32.28 An endoscopic view demonstrating the posterior (neural) compartment of the right pterygopalatine fossa (00 lens endoscope). The principal vascular structures and fat have been removed. (Image courtesy of Professor C Snyderman, University of Pittsburgh Medical Center.)

Artery of the pterygoid canal

The artery of the pterygoid canal (Vidian artery) usually arises as a branch of the distal part of the maxillary artery or sometimes as a branch of the petrous segment of the internal carotid artery. It passes through the pterygoid canal and anastomoses with the pharyngeal, ethmoidal and sphenopalatine arteries in the pterygopalatine fossa and with the ascending pharyngeal, accessory meningeal, ascending palatine and descending palatine arteries in the oropharynx and around the pharyngotympanic tube. Through these complex anastomoses, the artery of the pterygoid canal contributes to the supply of part of the pharyngotympanic tube, the tympanic cavity and the upper part of the pharynx. It may also anastomose with the artery of the foramen rotundum and so communicate with branches of the cavernous portion of the internal carotid artery.

Pharyngeal artery

The pharyngeal branch of the maxillary artery passes through the palatovaginal canal, accompanying the nerve of the same name, and is distributed to the mucosa of the nasal roof, nasopharynx, sphenoidal air sinus and pharyngotympanic tube.

Greater (descending) palatine artery

The greater palatine artery leaves the pterygopalatine fossa through the greater (anterior) palatine canal, within which it gives off two or three lesser palatine arteries. The greater palatine artery supplies the inferior meatus of the nose, then passes on to the roof of the hard palate at the greater (anterior) palatine foramen and runs forwards to supply the hard palate and the palatal gingivae of the maxillary teeth. It gives off a branch that runs up into the incisive canal to anastomose with the nasopalatine artery, and so contribute to the arterial supply of the nasal septum. The lesser palatine arteries emerge on to the palate through the lesser (posterior) palatine foramen, or foramina, and supply the soft palate.

Sphenopalatine artery

The sphenopalatine artery and the greater palatine artery are the terminal branches of the maxillary artery. The sphenopalatine artery is the principal artery supplying the mucosa of the nose. It enters the nasal cavity through the sphenopalatine foramen posterior to the superior meatus. From here, its posterior lateral nasal branches ramify over the conchae and meatuses, anastomosing with the ethmoidal arteries and nasal branches of the greater palatine artery to supply the frontal, maxillary, ethmoidal and sphenoidal air sinuses. The sphenopalatine artery next crosses anteriorly on the inferior surface of the sphenoid and ends on the nasal septum in a series of posterior septal branches that anastomose with the ethmoidal arteries (see Fig. 33.9).

Spatial arrangement of the neurovascular contents in the pterygopalatine fossa

The maxillary artery and its branches are located in a plane anterior to the maxillary nerve, pterygopalatine ganglion and the nerve of the pterygoid canal. This spatial arrangement is particularly relevant in endoscopic approaches to the pterygopalatine fossa (Video 32.2). The veins accompanying the third (pterygopalatine) part of the maxillary artery, in common with those accompanying the second (pterygoid) part, are in the form of a venous plexus. Profuse haemorrhage may be encountered from this pterygopalatine venous plexus.

Maxillary nerve

The maxillary division of the trigeminal nerve is wholly sensory (Fig. 32.30). It leaves the skull via the foramen rotundum and enters the upper part of the pterygopalatine fossa, from which all its extracranial branches are derived. The meningeal branch is given off in the middle fossa. Crossing the pterygopalatine fossa, the nerve gives off two large ganglionic branches that contain fibres destined for the nose, palate and pharynx, and these pass through the pterygopalatine ganglion without synapsing. It then inclines sharply laterally on the posterior surface of the orbital process of the palatine bone and on the upper part of the posterior surface of the maxilla in the inferior orbital fissure (which is continuous posteriorly with the pterygopalatine fossa); it lies outside the orbital periosteum, and gives off its zygomatic and then posterior superior alveolar branches. About halfway between the orbital apex and the orbital rim, the maxillary nerve turns medially to enter the infra-orbital canal as the infraorbital nerve. The subsequent course of the maxillary nerve is described on page 502.

The maxillary nerve gives off many of its branches in the pterygopalatine fossa. They can be subdivided into those that come directly from the nerve, and those that are associated with the pterygopalatine parasympathetic ganglion. Named branches from the main trunk are meningeal, ganglionic, zygomatic, posterior, middle and anterior superior alveolar and infraorbital nerves. Named branches from the pterygopalatine ganglion are orbital, nasopalatine, posterior superior nasal, greater (anterior) palatine, lesser (posterior) palatine and pharyngeal.

Meningeal nerve

The meningeal branch of the maxillary nerve arises within the middle cranial fossa and runs with the middle meningeal vessels. It contributes to the innervation of the dura mater.

Ganglionic branches

There are usually two ganglionic branches that connect the maxillary nerve to the pterygopalatine ganglion.

Zygomatic nerve

The zygomatic branch of the maxillary nerve leaves the pterygopalatine fossa through the inferior orbital fissure together with the maxillary nerve. Its subsequent course is described on page 502.

Posterior superior alveolar nerve

The posterior superior alveolar nerve leaves the maxillary nerve in the pterygopalatine fossa. Its subsequent course and distribution are described on page 526.

Infraorbital nerve

The infraorbital nerve can be regarded as the terminal branch of the maxillary nerve. It leaves the pterygopalatine fossa to enter the orbit at the inferior orbital fissure, and its subsequent course and distribution are described on page 502.

Orbital branches

Fine orbital branches enter the orbit through the inferior orbital fissure and supply orbital periosteum. Some fibres also pass through the posterior ethmoidal foramen to supply the sphenoidal and ethmoidal sinuses. The orbital branches are believed to join branches of the internal carotid nerve to form a 'retro-orbital' plexus, from which orbital structures such as the lacrimal gland and orbitalis receive an autonomic innervation (Ruskell 2004).

Nasopalatine nerve

The nasopalatine nerve leaves the pterygopalatine fossa through the sphenopalatine foramen and enters the nasal cavity. It passes across the cavity to the back of the nasal septum, runs downwards and forwards on the septum in a groove in the vomer, and then turns down through the incisive fossa in the anterior part of the hard palate to enter the roof of the mouth (see Fig. 33.11). When an anterior and a posterior incisive foramen exist in this fossa, the left nasopalatine nerve passes through the anterior foramen, and the right nerve passes through the posterior foramen. The nasopalatine nerve supplies the lower part of the nasal septum and the anterior part of the hard palate, where it communicates with the greater palatine nerve. The nasopalatine artery, a branch of the maxillary artery, accompanies the nasopalatine nerve, ending in the incisive canal, where it anastomoses with greater palatine artery and the septal branch of the superior labial artery.

Posterior superior nasal nerves (lateral and medial)

The posterior superior alveolar nerves enter the back of the nasal cavity through the sphenopalatine foramen. Lateral posterior superior nasal nerves (about six) innervate the mucosa lining the posterior part of the superior and middle nasal conchae and the posterior ethmoidal sinuses. Two or three medial posterior superior nasal nerves cross the nasal roof below the opening of the sphenoidal sinus to supply the mucosa of the posterior part of the roof and of the nasal septum.

Palatine nerves (greater and lesser)

The greater and lesser palatine nerves pass downwards from the pterygopalatine ganglion through the greater palatine canal. The greater palatine nerve descends through the greater palatine canal, emerges on the hard palate from the greater palatine foramen and runs forwards in a groove on the inferior surface of the bony palate almost to the incisive teeth. It supplies the gingivae, mucosa and glands of the hard palate and also communicates with the terminal filaments of the nasopalatine nerve. In the greater palatine canal, it gives off posterior inferior nasal branches that emerge through the perpendicular plate of the palatine bone and ramify over the inferior nasal concha and walls of the middle

The internal carotid artery–external carotid artery communication via the artery of the pterygoid canal is one of a number of collateral vascular pathways that exist between the external carotid artery and branches of the petrous and intracavernous segments of the internal carotid artery. Albeit small and difficult to demonstrate angiographically, these vascular connections may nevertheless provide the arterial supply for arteriovenous malformations, dural-based fistulae and skull base tumours, and may also compensate for occlusion of the internal carotid artery. They also pose a risk of inadvertent embolization of the brain parenchyma during interventional angiography of the external carotid artery system.

In view of the extensive communications between the artery of the pterygoid canal and the highly vascular nasopharyngeal mucosa, for example, embolization of the external carotid artery branches may result in incomplete obliteration of a vascular nasopharyngeal lesion when the artery of the pterygoid canal arises from the internal carotid artery. Conversely, the collateral circulation provided by the artery of the pterygoid canal may play a role in maintaining cerebral circulation if there is stenosis of the petrous internal carotid artery proximal to the point where the artery of the pterygoid canal joins it; the flow in the post-stenotic segment of the internal carotid artery will be supplemented from the external carotid artery via the artery of the pterygoid canal (**Fig. 32.29**) (Tubbs et al 2007, Takeuchi et al 2005, Allen et al 2005).

As the greater palatine artery runs the length of the hard palate, it permits axial mucoperiosteal flaps to be raised, if necessary bilaterally, for the reconstruction of local defects. It is also possible to pedicle the entire mucoperiosteum of the hard palate on a single greater palatine artery. The vascular supply to the denuded palatal bone is sufficient, presumably from its nasal surface, to permit regeneration of the palatal soft tissue. Contracture of the regenerated tissue does not occur because the defect lies over the palatal bone (Gullane and Arena 1977, Gullane and Arena 1985, Salins and Benjamin 2009).

Ligation of the sphenopalatine artery may be necessary in the control of refractory epistaxis. It is usually located endoscopically at the posterior/inferior end of the middle turbinate. Multiple branches may be encountered. The sphenopalatine artery provides the axial blood supply to the nasal septal mucosal flap (nasoseptal flap) used in endoscopic skull-base repair, a reflection of its contribution to the vascular supply to the nasal cavity (Hadad et al 2006).

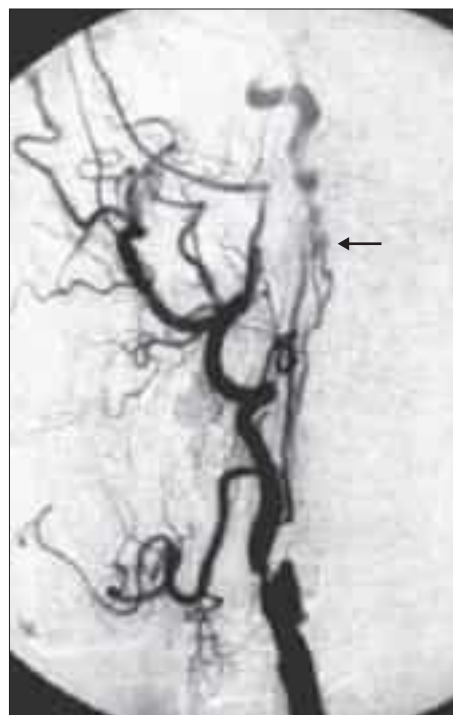


Fig. 32.29 A right common carotid angiogram, left anterior oblique view. The ascending pharyngeal artery provides a collateral blood flow to the petrous internal carotid artery via the artery of the pterygoid canal (arrow). (With permission from Takeuchi M, Kuwayama N, Kubo M, et al. Vidian artery as a collateral channel between the external and occluded internal carotid arteries. *Neuro Med Chir* (Tokyo) 2005;45:470–471.)

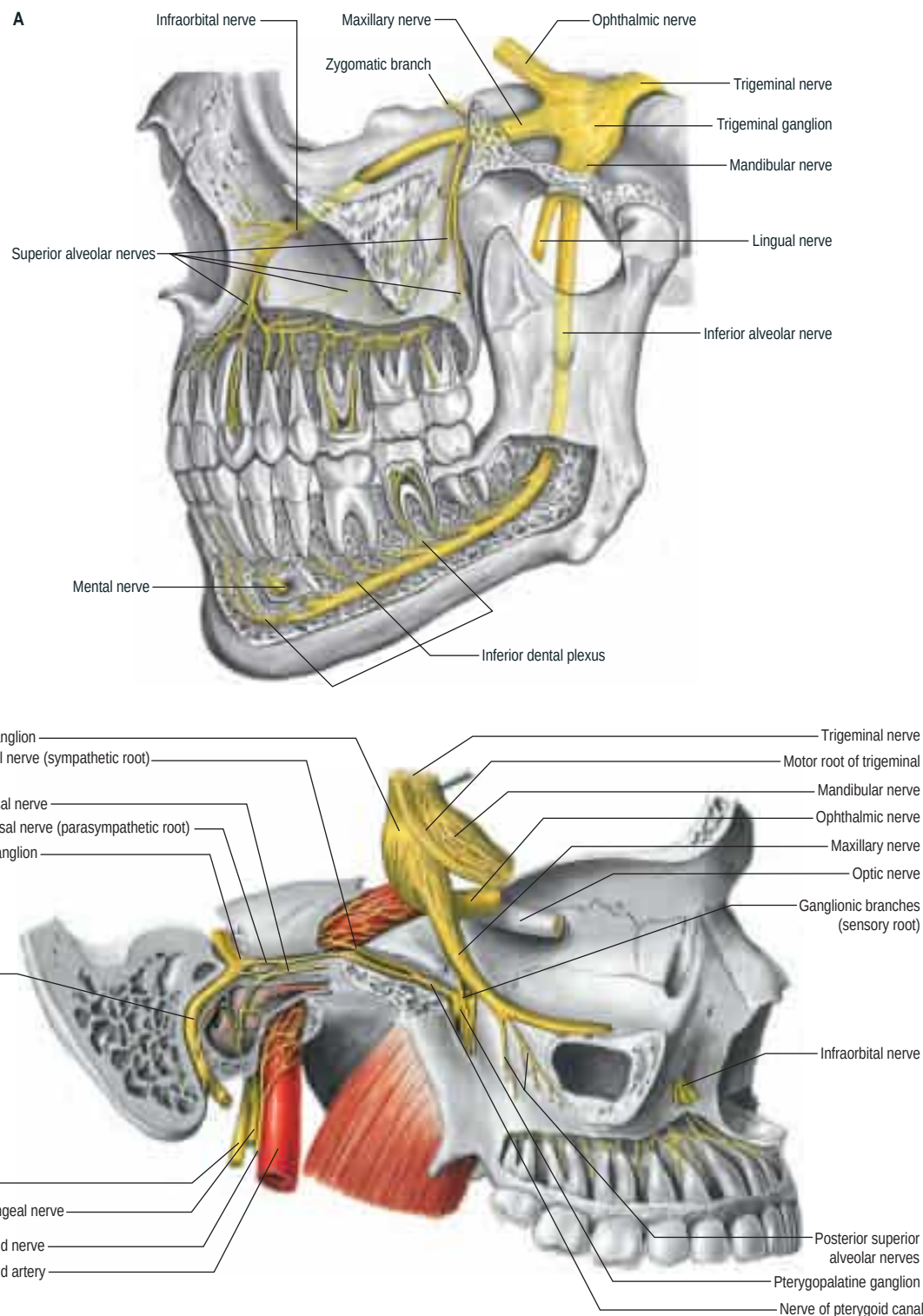


Fig. 32.30 A, Maxillary and mandibular nerves. **B**, The nerves of the pterygoid canal and the pterygopalatine ganglion. The trigeminal ganglion has been lifted forwards out of the trigeminal impression. Note the sympathetic plexus on the internal carotid artery. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

and inferior meatuses. As it leaves the greater palatine canal, it gives off branches that are distributed to both surfaces of the adjacent part of the soft palate (see Fig. 31.3).

The lesser (middle and posterior) palatine nerves are much smaller than the greater palatine nerve. They descend through the greater palatine canal, from which they diverge low down to emerge through the lesser palatine foramina in the tubercle or pyramidal process of the palatine bone. They innervate the uvula, tonsil and soft palate.

Fibres conveying taste impulses from the palate probably pass via the palatine nerves to the pterygopalatine ganglion. They pass through the ganglion without synapsing and leave via the greater petrosal nerve. Their cell bodies are located in the facial ganglion and their central processes pass via the sensory root of the facial nerve (nervus intermedius) to the gustatory nucleus in the nucleus of the tractus solitarius.

Pharyngeal nerve

The pharyngeal branch of the maxillary nerve leaves the pterygopalatine ganglion posteriorly. It passes through the palatovaginal canal with the pharyngeal branch of the maxillary artery and supplies the mucosa of the nasopharynx behind the pharyngotympanic tube.

Neural anastomoses with the maxillary nerve



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Pterygopalatine ganglion

The pterygopalatine ganglion, the largest of the peripheral parasympathetic ganglia, is well protected, deep in the pterygopalatine fossa.

Extensive anastomoses, which may be sensory-to-sensory, sensory-to-motor or motor-to-motor, exist between the lower cranial nerves and upper cervical nerves. Variable anastomoses exist between all three of the divisions of the trigeminal nerve with branches of the facial nerve. The majority of trigeminal nerve interconnections are with the facial nerve. Connections between the zygomaticotemporal branch of the maxillary nerve and the temporal branches of the facial nerve, and the zygomaticofacial branch of the maxillary nerve and the zygomatic branches of the facial nerve have been demonstrated ([Shoja et al 2014](#)). The connection between the zygomaticotemporal and lacrimal nerves in the orbit providing parasympathetic secretomotor fibres to the lacrimal gland, as was once thought, is less certain. It is more likely that the postganglionic orbital branches pass through the inferior orbital fissure and innervate the lacrimal gland directly ([Rusnell 2004](#)). It has been suggested that one possible explanation for these trigeminal–facial interconnections relates to the proprioceptive innervation of the facial musculature but the functional significance of some of the neural connections has yet to be determined.

Immediately anterior to the pterygoid canal, the ganglion lies below and medial to the foramen rotundum and the maxillary nerve, and lateral to the sphenopalatine foramen (see Fig. 32.30B). It is flattened and reddish-grey in colour, and lies just below the maxillary nerve as it crosses the pterygopalatine fossa. The majority of the 'branches' of the ganglion are connected with it morphologically but not functionally, because they are primarily sensory branches of the maxillary nerve. Thus they pass through the ganglion without synapsing and enter the maxillary nerve through its ganglionic branches; they convey some parasympathetic fibres to the palatine, pharyngeal and nasal mucous glands.

Preganglionic parasympathetic fibres destined for the pterygopalatine ganglion run initially in the greater petrosal branch of the facial nerve, and then in the nerve of the pterygoid canal (Vidian nerve), after the greater petrosal unites with the deep petrosal nerve. The nerve of the pterygoid canal enters the ganglion posteriorly. Postganglionic parasympathetic fibres leave the ganglion; some may join the maxillary nerve via a ganglionic branch and then travel via the zygomatic and

zygomaticotemporal branches of the maxillary nerve to the lacrimal gland, but most are thought to travel by orbital branches to the gland (see above). The parasympathetic supply to the pre-ocular part of the central artery of the retina is believed to be derived mainly from the pterygopalatine ganglion and is vasodilatory (Ch. 42). Preganglionic secretomotor fibres of uncertain origin also travel in the nerve of the pterygoid canal. They synapse in the pterygopalatine ganglion, and postganglionic fibres are distributed to palatine, pharyngeal and nasal mucous glands via palatine and nasal branches of the maxillary nerve.

Postganglionic sympathetic fibres pass through the ganglion without synapsing and supply blood vessels and orbitalis. They arise in the superior cervical ganglion and travel via the internal carotid plexus and deep petrosal nerve to enter the pterygopalatine ganglion within the nerve of the pterygoid canal.

General sensory fibres destined for distribution via orbital, nasopalatine, superior alveolar, palatine and pharyngeal branches of the maxillary division of the trigeminal nerve run through the ganglion without synapsing.



Bonus e-book images and videos

Fig. 32.2 The 'extended' infratemporal fossa.

Fig. 32.4 The stylohamular plane.

Fig. 32.8 The opposing direction of resorption ('atrophy') of the mandible and maxilla, resulting in increasing incongruity between the jaws.

Fig. 32.9 A bilateral fracture of an edentulous atrophic mandible.

Fig. 32.10 A CT image showing the lateral lingual foramina.

Fig. 32.11 A three-dimensional reformatted CT image showing an implant perforating the lingual cortical plate, leading to haemorrhage in the anterior floor of the mouth.

Fig. 32.12 A transverse section at the level of the hard palate demonstrating the cranial fascia and associated compartments.

Fig. 32.19 The arterial network of the human temporalis muscle.

Fig. 32.24 The foramen ovale, inferolateral view.

Fig. 32.25 Anastomoses of the auriculotemporal nerve with the temporofacial division of the facial nerve, and of the great auricular nerve with the cervicofacial division of the facial nerve in the parotid and periparotid regions.

Fig. 32.26 Ludwig's angina: spread of infection from an infected lower left third molar to the pterygomandibular space and thence into the submandibular, sublingual and submental spaces bilaterally.

Fig. 32.27 An endoscopic view demonstrating the anterior (vascular) compartment of the right pterygopalatine fossa (00 lens endoscope).

Fig. 32.28 An endoscopic view demonstrating the posterior (neural) compartment of the right pterygopalatine fossa (00 lens endoscope).

Fig. 32.29 A right common carotid angiogram, left anterior oblique view.

Video 32.1 Temporomandibular joint arthroscopy demonstrating intracapsular anatomy of the joint.

Video 32.2 Endoscopic anatomy of the infratemporal and pterygopalatine fossae.

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